

AN INVESTIGATION OF THE RADIATED SIGNALS
PRODUCED BY SMALL SPARKS ON POWER LINES

A Dissertation

by

WILLIAM LOUIS BEASLEY

Submitted to the Graduate College of
Texas A&M University in
partial fulfillment of the requirement for the degree of

DOCTOR OF PHILOSOPHY

January 1970

Major Subject: Electrical Engineering

AN INVESTIGATION OF THE RADIATED SIGNALS

PRODUCED BY SMALL SPARKS ON POWER LINES

A Dissertation

by

WILLIAM LOUIS BEASLEY

Approved as to style and content by:

John P. German

(Chairman of Committee)

W.B. Jones

(Head of Department)

George F. Kriebner Jr.

(Member)

Jack T. Kent

(Member)

Albert J. Perry

(Member)

John A. Skerison

(Member)

January 1970

ABSTRACT

An Investigation of the Radiated Signals Produced
by Small Sparks on Power Lines. (January 1970)

William Louis Beasley, B.S.,

Agricultural and Mechanical College of Texas;

M.E., Texas A&M University;

Directed by: Dr. John P. German

This study reports the writer's findings about the time domain and frequency domain characteristics of the electromagnetic interference produced by the small-dimension, low root-mean-square current sparks which exist between metallic items on electric power lines.

The physics of the discharge is discussed and an explanation is given for the very short formation time, the narrow pulse width, and the rapid rate of decay of the spark. A measured plot is shown of the time domain waveform of the conducted spark current. This waveform is then used to calculate the coefficients of a Fourier series in order to give a mathematical expression in the frequency domain.

The method of measuring the amplitude of the spark noise spectrum in terms of peak volts per unit bandwidth is described. A mathematical proof is given which shows the spectrum is independent of the period of the spark. An analytical equation is then developed which expresses the spark noise amplitude as a function of

frequency.

A model to represent the spark as a noise source is postulated. A description is given of the association of the spark with dipole radiators representing some typical pole-line hardware configurations. The resulting radiated electric field strength is calculated and compared to measured values. The postulated model characteristics are then shown to agree reasonably well with the measured characteristics of the noise source.

ACKNOWLEDGEMENTS

The writer is grateful for the many considerations extended to him by the members of his graduate committee: to the chairman, Dr. J. P. German, for his supervision and guidance, to Professor John Denison and Albert Druce for their advice and council, and to Professor J. T. Kent and Dr. George Huebner for serving on the committee. Especial thanks are extended to Dr. W. B. Jones, Jr. and Dr. G. D. Hallmark for their assistance and encouragement throughout the research period.

The writer is deeply grateful to his beloved wife, Vera Mae, for her abiding patience, understanding, and encouragement. Lastly, to the typist, his daughter Kassie, thanks is extended for typing this dissertation.

TABLE OF CONTENTS

Chapter		Page
I	INTRODUCTION	1
II	THEORY OF THE SPARK DISCHARGE	5
III	MEASUREMENTS OF THE TIME DOMAIN WAVEFORM OF THE SPARK	11
IV	FREQUENCY DOMAIN CHARACTERISTICS OF THE SPARK	27
V	THE ELECTROMAGNETIC RADIATION PRODUCED BY THE SPARK	48
VI	CONCLUSIONS	63
	BIBLIOGRAPHY	65
	APPENDICES	
A	COMPUTER PROGRAM	67
B	MEASURED VALUES	71
VITA		88

LIST OF TABLES

Table No.		Page
V-1	List of Dipole Lengths, Center Capacitance, and Resonant Frequencies . .	50

LIST OF FIGURES

Figure No.		Page
3-1	Plot of Minimum Voltage or Minimum Electric Field Necessary to Produce a Spark versus Gap Length.	12
3-2	Plot of Electrode Voltage versus Arc Current for Two Gap Lengths.	14
3-3	Schematic Diagram of Static Characteristic Measurements Apparatus.	16
3-4	Plot of Voltage versus Current during a typical Spark Pulse.	18
3-5	Plot of Amplitude versus Time of the Spark Current.	20
3-6	Coincident Plots of Current Pulses vs. Time and Electrode Voltage vs. Time. . . .	22
3-7	Coincident Plots of Spark Current vs. Time and Electrode Voltage vs. Time for a 0.030 inch Gap.	24
3-8	Plot of Spark Current or Arc Current and Source Voltage vs. Time.	25
4-1	Schematic Diagram of Frequency Domain Measurement Apparatus.	32
4-2	Plot of Noise Voltage Amplitude Calculated from Time Domain Waveform.	38
4-3	Plot of $V(f)$ Calculated from Equation (IV-20).	44
4-4	Preliminary Model of the Spark.	45
4-5	Final Model of the Spark.	46
5-1	Schematic Diagram of Apparatus Used to Measure Radiated Field Strength. . . .	52

LIST OF FIGURES (cont'd.)

Figure No.		Page
5-2	Measured Radiated Electric Field Strength 21.35 meters from dipole. . . .	54
5-3	Calculated Radiated Electric Field Strength 21.35 meters from dipole. . . .	58
5-4	Comparison of Measured and Calculated Electric Field Strength.	59

CHAPTER I

INTRODUCTION

The transmission of large amounts of information from one location to another is one aspect of the current technology explosion. Often, the most practical method of transporting information is an electromagnetic wave radiated through space. The information contained in the radiated wave is in the form of a variation in the amplitude, phase, or frequency parameter of a high frequency sinusoid.

It is desirable that the radiated wave (which represents the transmitted information) arrive at its destination with the parameter unchanged. However, the parameter is often modified resulting in the reception of contaminated information. That the original form of the message is changed during transmission is evident when one considers the distorted television picture, the unintelligible news broadcast, or inaccurate telemetered data. This contamination of the message is usually caused by the addition of undesired waveforms to the radiated wave.

The term noise may be applied to all waveforms present other than the desired message. These undesired waveforms have many sources, including natural phenomena such as lightning discharges in the atmosphere or the random fluctuations of electrons in conductors, transistors and vacuum tubes. Other sources of noise are man-made and may be another message signal or may be an unintention-

The citations on the following pages follow the style of the IEEE TRANSACTIONS on POWER APPARATUS AND SYSTEMS.

tional product of the normal use of electrical equipment. The noise considered in this investigation is of the last type named and is specifically described below.

The communication engineer often finds a message, as represented by a variation in a parameter of a sinusoidal electromagnetic wave, contaminated by another signal of the same type. This interference often may be controlled by adequate filtering, frequency shifting, or proper geographical spacing. Conversely, the broadband noise signal produced by quasi-periodic impulsive waveforms with very short rise and fall times may be very difficult to control. This noise cannot be filtered without losing part of the desired message, it occupies a broad band of frequencies and its source is often situated in the right location to contaminate the message.

The small current, small dimension spark is a source of this broadband noise and may be found on electric power distribution and transmission lines whose potentials are above a few kilovolts (kV). This small spark is rarely found on the low voltage secondary of a distribution system. The reason for this is that there exists a minimum sparking potential of about 300 volts for air [1]. Under normal conditions, a spark will not form in a gap of any length when the potential is below this minimum value.

Most of the small spark noise sources are found on older distribution and transmission lines with a difference of potential of 4 kV or more. The spark usually occurs across an air gap between items of pole line hardware. These air gaps may be due to improperly

placed guy wires or ground wires, or may result from loose metallic items on the pole. Gaps are often caused by insulating oxides forming on metallic surfaces of bell-type insulators. An additional requirement for spark formation is that points of large potential difference must be connected to the gap electrodes through a high impedance path. This path may be the moist interior of the supporting wood members, conductive salts on insulator surfaces or capacitance between metal components.

The relaxation type oscillation which results from certain combinations of the conditions described above consists of the spark gap alternately becoming conducting and nonconducting. The very high rate of change of the conductance of the spark gap as the spark forms and relaxes produces a current waveform whose amplitude spectrum extends into the ultra high frequency band.

Preliminary calculations based on a time domain graph of the current waveform of the spark indicate that the amplitude spectrum will extend well into the gigahertz (GHz) region.

Pakala, et. al., [2] measured an amplitude spectrum of the conducted current in an artificial gap on a power line. He reported that the spectrum was flat to 30 megahertz (MHz), decreased rapidly to 200 MHz and fluctuated markedly for higher frequencies. He attributed these fluctuations to impedance variations of the test circuit. Similar observations were made during the early part of the investigation reported herein, indicating that test circuits must be carefully designed.

Skomal [3] believes that power lines with potential differences of 4 kV or more may be large contributors to radio interference for frequencies up to 100 MHz. German [4] reported measurements of noise signals originating from gap discharges with components whose frequencies extend to one GHz.

The spark described above is associated with some metallic hardware on the power line and thus represents a source of radio frequency (RF) energy attached to a radiating system. The frequency domain characteristics of the radiated signal produced by this association of a source with a radiating element will be investigated in this research.

CHAPTER II

THEORY OF THE SPARK DISCHARGE

This chapter deals chiefly with an investigation of the physics of the discharge occurring between conductors separated a few tens of thousandths of an inch and immersed in air at normal atmospheric pressure. An explanation will be given for the very short time required for the formation and relaxation of the discharge.

The Beginning Discharge

The resistance between a pair of conductors immersed in air at atmospheric pressure is very high if the potential between them is small. Air is thus an excellent insulator and the current existing between conductors is extremely small as long as the electric field strength is moderate.

As the potential between the conductors is increased to higher levels, one observes a sudden and marked change in the insulating property of air. The resistance between the two conductors decreases to a low value, the current increases many orders of magnitude and air becomes a relatively good conductor of electricity.

The sudden change described above is usually noisy, luminous and explosive in nature. The transition may end in a continuous current called an arc if the power available is large and the electrical circuit resistance is low [5]. The transition may begin as outlined above but a high circuit resistance may prevent the

formation of a stable arc. In either example described above the transition process is usually called a spark. However, laboratory measurements made in this study revealed that a small current continuous arc with a conduction current in the order of milliamperes (mA) does not produce radio frequencies (RF) with measurable amplitudes and will not be considered in this investigation.

The Quasi-periodic Spark

The spark which does not turn into an arc but terminates as it began with the air returning to its high resistance and good insulating property will produce electromagnetic waves which extend into ultra high frequency (UHF) range [6]. This spark is usually repetitive with random periods between pulses and will continue as long as the high potential exists between the conductors. This is the type of discharge which does produce RF and will be investigated in this research.

The mechanism by which breakdown occurs has been exhaustively investigated by such scientists as Sir J. J. Thompson, J. S. Townsend and Ernest Rutherford and reference to their work will be made.

The breakdown process begins with one or more free electrons existing in the space between electrodes. The electric field due to the potential between electrodes will accelerate each electron and impart kinetic energy to it. If the electron acquires sufficient kinetic energy it may collide with and ionize a molecule of a gas, resulting in one or more additional free electrons and

a positive ion. These free electrons repeat the ionization process described above, resulting in an avalanche in which the ionization by collision grows rapidly. This ionization process is the cause of the rapid decrease in the resistance of the air gap between the electrodes.

The effect of the positive ion on the ionization process may be determined qualitatively by investigating the relative energies acquired by an electron and a positively ionized molecule. The electron and singly ionized molecule have a charge of the same magnitude and will experience the same magnitude of force when placed in the same electric field. The energy acquired by either charged body may be written

$$E = \frac{1}{2}mv^2 \quad (\text{II-1})$$

where E is energy in joules, m is mass in kilograms and v is velocity in meters per second. A body being subjected to a constant force will react with a constant acceleration and one may write

$$v = at \quad \text{or} \quad v = \frac{f}{m}t \quad (\text{II-2})$$

where a is acceleration in meters per second per second, f is force in newtons and t is time in seconds. One may substitute the expression for velocity into the expression for energy as follows:

$$E = \frac{1}{2}m\left(\frac{f}{m}t\right)^2 = \frac{1}{2} \frac{f^2 t^2}{m} \quad (\text{II-3})$$

One observes that for a constant force and a fixed time interval the energy acquired by a body is inversely proportional to its mass. Since the mass of the ionized molecule is several thousand times that of the electron, the kinetic energy which the positive ion acquires is many orders of magnitude smaller than that of the electron. As a result, the direct effect of the positive ion will be disregarded in the avalanche ionization process.

Theoretical investigations which include the mean free path of the electron and its acceleration due to the electric field fail by an order of magnitude to predict the short time required for the spark to form a conducting path between the electrodes. [7]. This short time can be explained by successive photo-ionizations of gas molecules some distance away from the primary avalanche. An additional factor is that the electric field is increased by positive ions remaining from earlier avalanches [8].

A better understanding of the photoionization process may be obtained by further consideration of the primary avalanche. Photons, the visible aspect of the spark as described in the first part of this chapter, are produced during this avalanche either as a part of the disassociation of the electrons and ions or by their recombination into neutral molecules. These photons, moving with the velocity of light, are capable of removing electrons from

other neutral molecules and may supply free electrons almost simultaneously at many places in the space between the electrodes. Each of the free electrons produced by the photons may now be the cause of new avalanches distributed throughout the space between the electrodes. The result could be described as an avalanche of avalanches, causing a large supply of charge carriers to become available in a very short period of time. The time required for the formation of the discharge path, as will be demonstrated by plots in a later chapter, is about one nanosecond (ns) and is very nearly independent of path length for spacings from 0.01 inch (0.254 mm) to 0.2 inch (5.08 mm).

The length of time required for the discharge to decay may be inferred by considering the small distributed capacitance associated with the electrodes between which the discharge occurs and by noting the low value of resistance which exists across the discharge path when the current in the discharge reaches a maximum. If one assumes a few picofarads of capacitance and a few hundred ohms of resistance, one obtains a discharge time constant (RC) of about one ns. A fall time of this magnitude will be demonstrated by plots in the next chapter.

The short rise time of the formation of the spark, the short fall time of the relaxation of the discharge and the narrow width of the pulse of current suggest that this time domain waveform will have a frequency domain amplitude spectrum extending into the higher frequencies. The time domain waveform of the spark

will be considered in the next chapter preliminary to its frequency domain investigation.

CHAPTER III

MEASUREMENTS OF THE TIME DOMAIN

WAVEFORM OF THE SPARK

In this chapter the time domain wave form and its period will be described and evaluated. A description will be given of the method used to choose the parameters which represent a typical power line spark and the time domain characteristics will be shown by plots.

The Discharge Path Parameters

On power lines, the length of the gaps across which sparks occur and the type of electrode material varies from one location to another. Measurements of conduction currents in this research verified the work of earlier investigations [9], [10] that the type of electrode material has no measurable effect on the breakdown voltage or the discharge current. Since the type material was not a factor, electrodes of cast steel, 0.065 inch (1.65 mm) in diameter were chosen for the experimental work of this study. They were mounted in a support which permitted rapid and accurate adjustment of their spacing.

Distribution system line voltages usually are about 4 kV on the older systems, but may be higher on the newer ones. Figure 3-1 shows that with a gap length of 0.030 inch (0.762 mm), the lowest distribution voltage normally used could produce a spark.

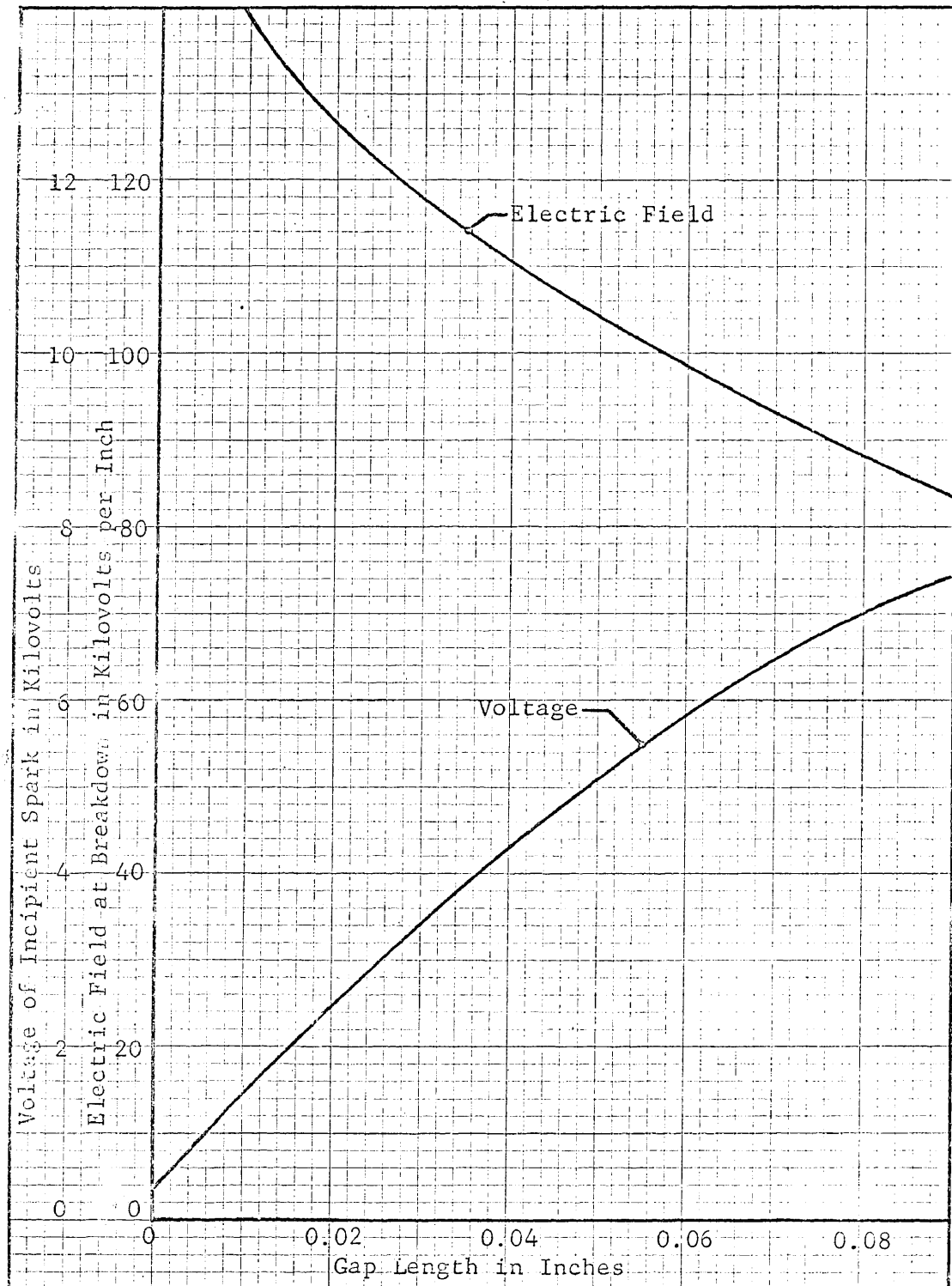


Fig. 3-1 Plot of Minimum Voltage or Minimum Electric Field Necessary to Produce a Spark versus Gap Length.

This spacing was selected as being representative of a typical spark gap to be found on distribution lines and was used during most of the measurements of this study.

In order for the spark to terminate without forming a continuous arc, a high resistance path must exist between the electrodes and the points of high potential. The size of this resistance may be obtained by using the information shown in Figure 3-2. This plot reveals that for a spacing of 0.030 inch (0.762 mm) between electrodes and a source voltage of 4 kV, a series resistance of 10 megohms will not allow sufficient current to exist to support a continuous arc.

The parameters described above were used while making most of the measurements recorded during this study. They were used because they produced discharges similar to the sparks found on distribution systems.

The Spark Excited by a dc Source

The preliminary measurements in this study were made with an adjustable dc source in order to obtain the spark characteristics as a function of voltage. Subsequent measurements were made with a sinusoidal 60 hertz (Hz) source similar to the voltages on power lines.

In order to determine the time domain characteristics of the spark, the electrodes were spaced 0.030 inch (0.762 mm) and connected to a variable dc source through a 10 megohm resistor.

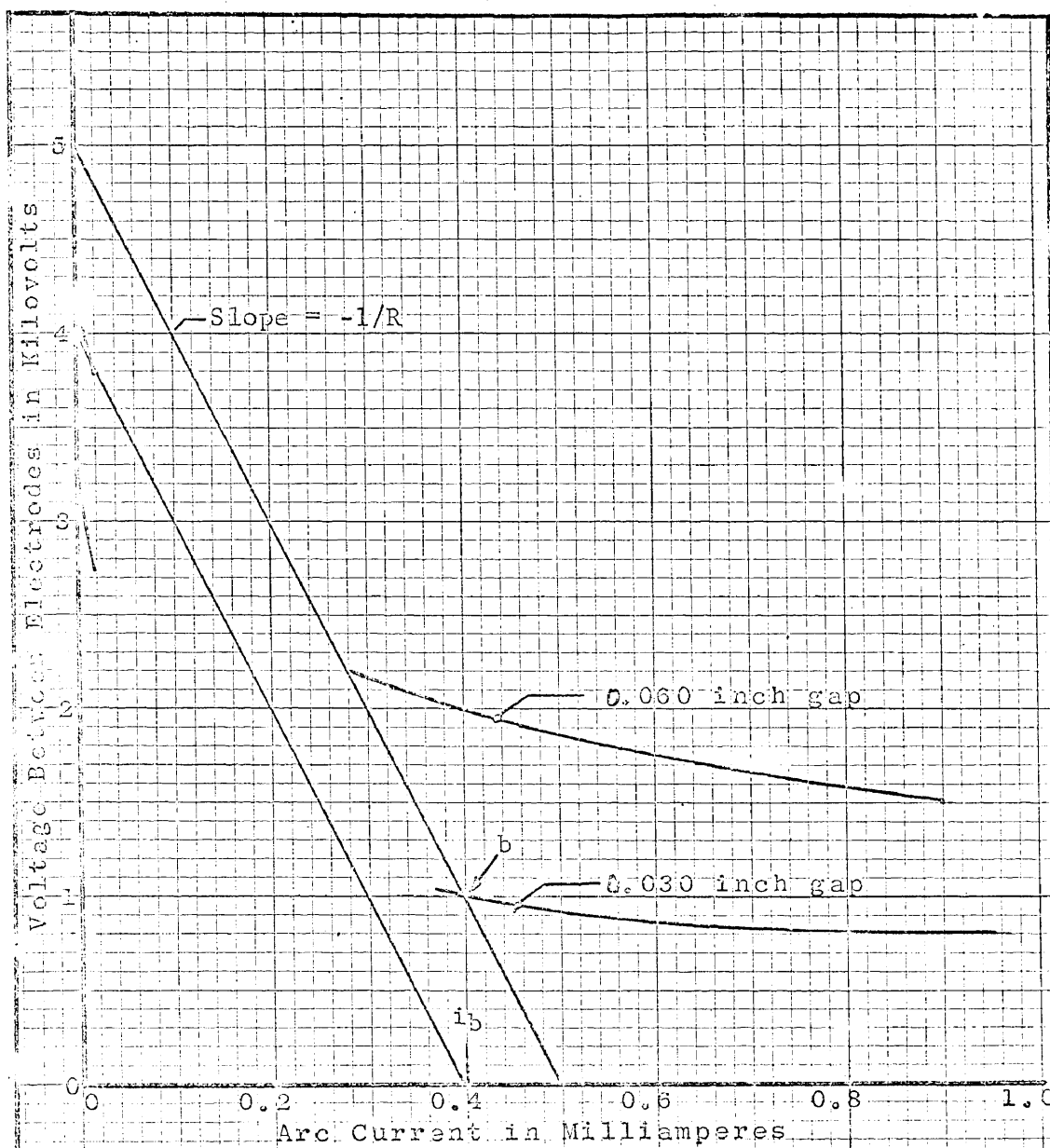


Fig. 3-2 Plot of Electrode Voltage versus Arc Current for Two Gap Lengths.

In the experimental part of this study, the spark current was measured in terms of a voltage across a 50 ohm resistor which was in series with and very near one electrode. The 50 ohm resistor, whose impedance was measured and found to be almost purely resistive up to 800 MHz, was connected through a 50 ohm co-axial cable to a sampling oscilloscope. A schematic diagram of the circuit described above is shown in Figure 3-3. Most of the measurements were of the direct type, but the arc voltage across the electrodes was obtained indirectly by subtracting the dc voltage V_R across the 10 megohm resistor R from the source voltage V_s .

The breakdown voltage V_b or minimum sparking potential for the 0.030 inch (0.762 mm) gap is about 3 kV and the minimum current which will sustain an arc is 0.4 mA. These values of voltage and current are important references which will be used in the following paragraph.

The steady-state solution for the current in the nonlinear series circuit may be obtained graphically from Figure 3-2 by observing the intersection of the spark characteristics with the $-\frac{1}{R}$ line, where R is the series resistance in ohms. For example, with a source voltage $V_s = 5\text{kV}$ and $R = 10^7$ ohms, the intersection of the two curves is at point b (Figure 3-2) with a continuous current equal to i_b . This is the steady arc previously described and does not produce RF. This case is of no further interest in this study.

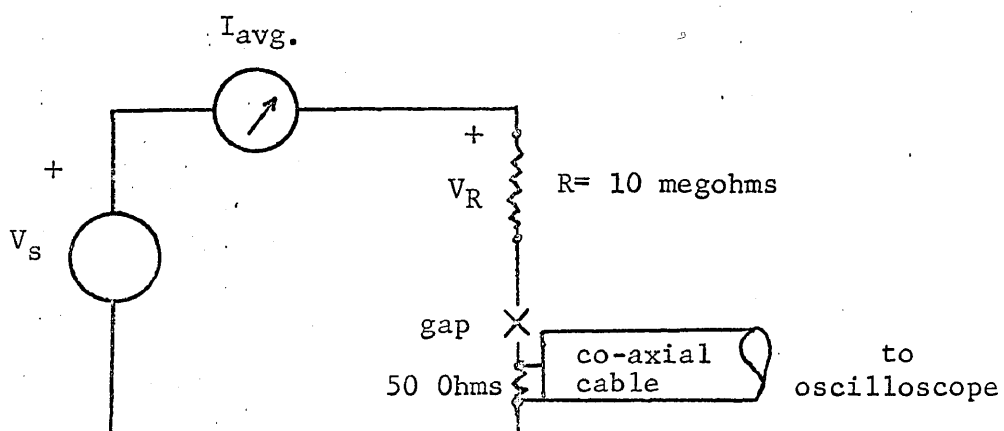


Fig.3-3 Schematic Diagram of Static Characteristic Measurements Apparatus.

Now consider the case when V_s is lowered so that the $-\frac{1}{R}$ line no longer intersects the spark characteristic curve but is still above the breakdown voltage. An incipient spark could not become an arc because a sustaining current could not exist. An apparent dilemma has been created in that the value of V_s indicates that a spark should occur but the current is insufficient to sustain it. The dilemma is only apparent and disappears when one adds the capacitance which exists because of the proximity of the electrodes. The addition of the capacitance of the circuit permits a relaxation type oscillation to occur in which the discharge current plotted versus voltage with time as a parameter is shown in Figure 3-4. The spark begins at point t_a when the voltage across the electrodes is at or above the breakdown value and there is an electron available to start the avalanche. As the spark path forms, the current rises to a maximum of several amperes and then decays toward zero as the stored energy is expended. The ionized path then relaxes because of the high series resistance and the voltage across the gap starts to increase to the source voltage with a time constant determined by the series resistance and the associated capacitance. The time required for the current to traverse the path from t_a to t_b to t_c is about 2 nanoseconds (ns) while the time required for the voltage to return to point t_a is a few tenths of a millisecond (ms).

The time domain waveform, as obtained by a Tektronix

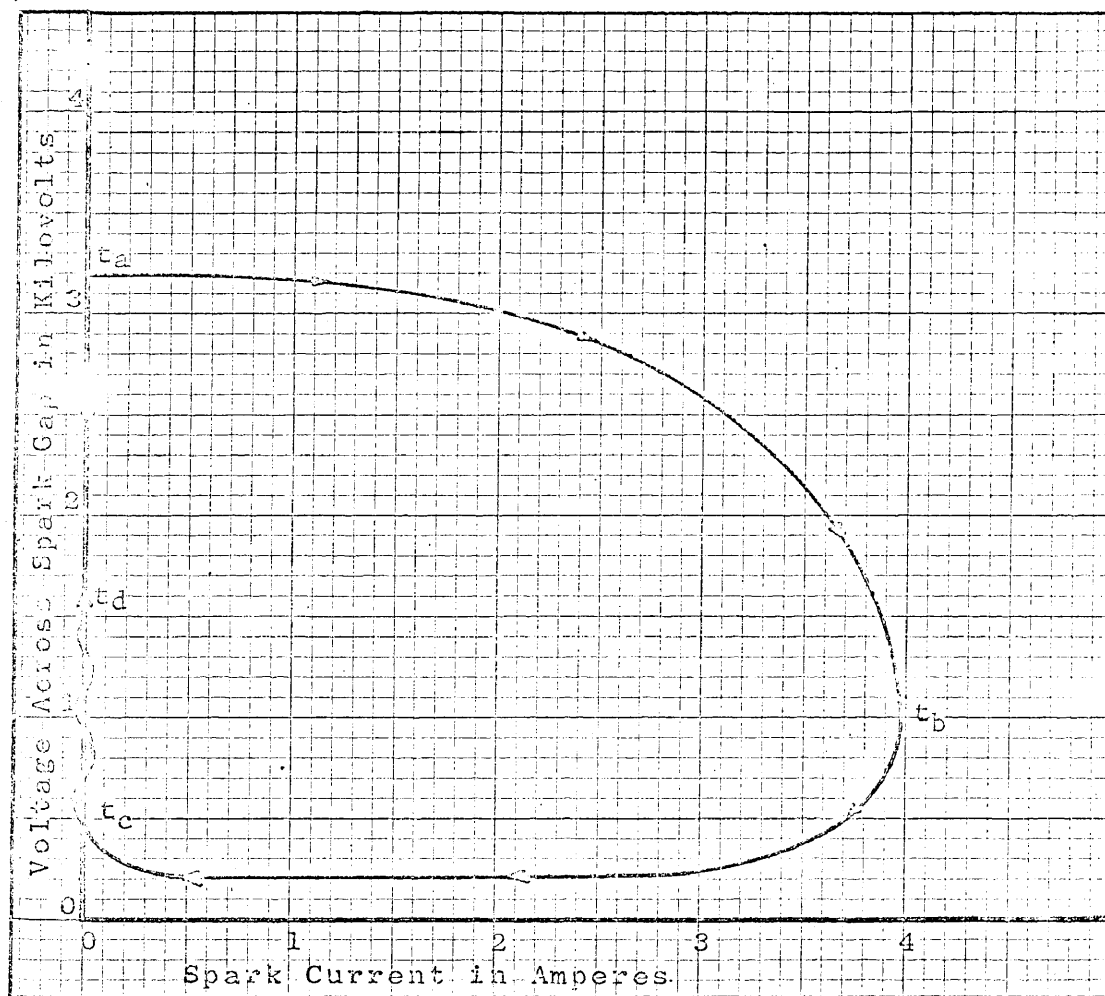


Fig. 3-4 Plot of Voltage versus Current during a typical Spark Pulse.

oscilloscope model 564, a sampling vertical amplifier model 3S76 and a sampling sweep model 3T77, is shown in Figure 3-5. The current was measured by observing the voltage across the previously described 50 ohm resistor in series with one electrode. The 50 ohm resistor was attached to a coaxial transmission line whose characteristic impedance was 50 ohms, resulting in an equivalent resistance, $R_{eq.}$, of 25 ohms. Thirty dB of attenuation was inserted in the coaxial cable (10 dB at the electrode and 20 dB at the oscilloscope) to perform the dual task of attenuating the very strong signal and improving the impedance match at each end of the cable.

This waveform was quasi-periodic, as will be described below, with an average period very long compared to the pulse width. The pulse was about 2 ns in width and the average period was about 10^5 times the pulse width. The trigger circuit of the sampling oscilloscope operated on the leading edge of each pulse so that a stationary pattern appeared on the cathode ray tube. The vertical and horizontal plug-in units of the oscilloscope supplied voltages to drive an x-y plotter for a permanent record.

The waveform on the oscilloscope did not change as the source voltage was varied from the minimum sparking potential to the maximum before the spark became a steady arc. This indicated there were sufficient free electrons in the gap so that the discharge began at about the same voltage each time,

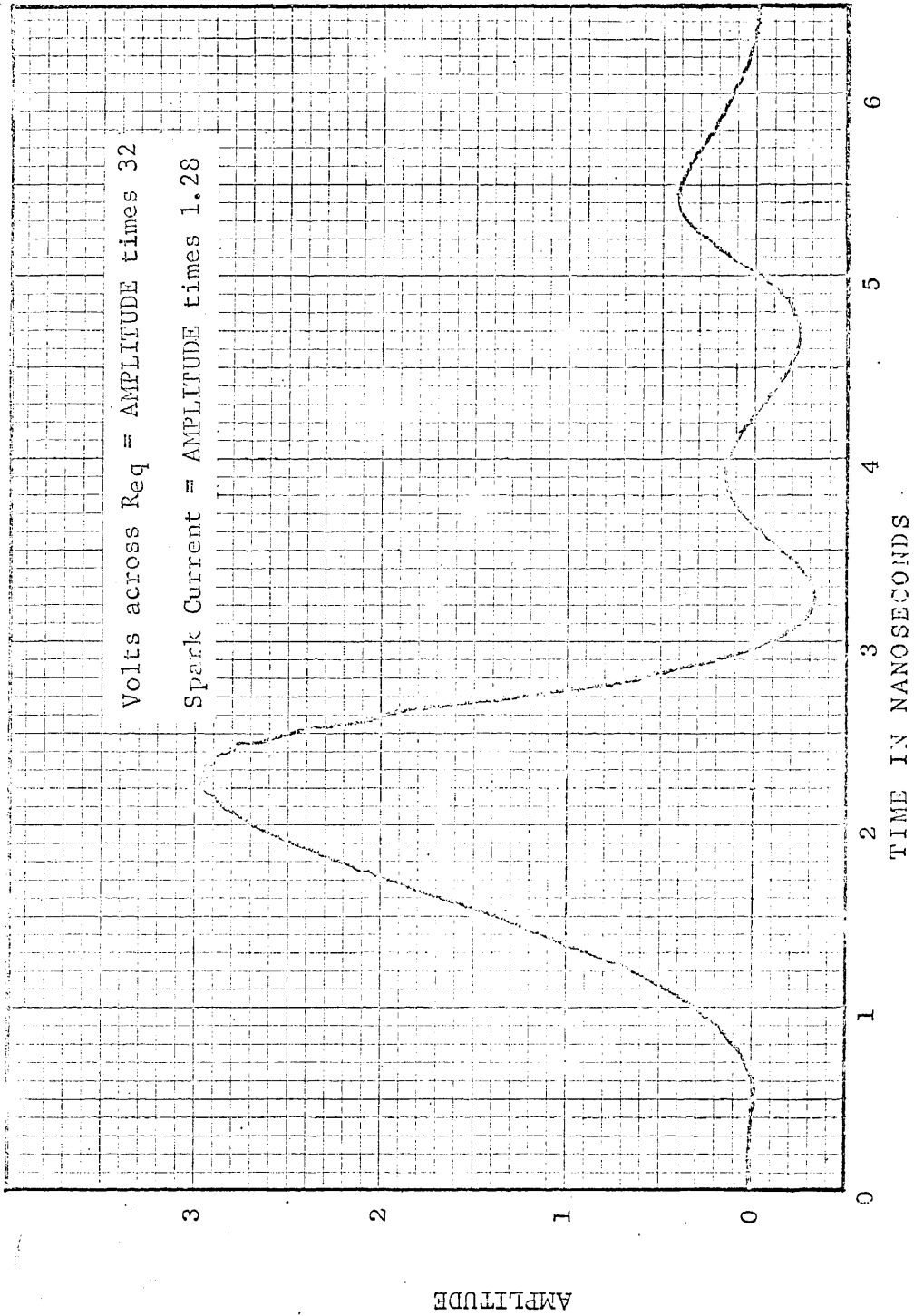


Fig. 3-5 Plot of Amplitude versus Time of the Spark Current.

resulting in identical pulses of current in each spark.

However, the average period changed, becoming shorter as V_s increased, because the exponential change of the voltage across the electrodes was increasing at a higher rate toward a higher steady state value. The quasi-periodic nature of the measured current pulses and the estimated voltage waveform are plotted with coincident time scales in Figure 3-6.

The peak value of the spark current may be calculated by using the information contained in Figure 3-5, the size of the equivalent series resistor (25Ω), and the amount of attenuation in the co-axial line to the oscilloscope. Thirty dB of attenuation corresponds to a reduction of a voltage amplitude by a factor of 0.0313, or the voltage across $R_{eq.}$ was equal to that measured by the oscilloscope multiplied by the reciprocal of 0.0313, or 32. The peak current may be calculated

$$i_p = \frac{32 \times v_p}{R_{eq.}}$$

where i_p is the peak current, v_p is the peak voltage recorded from the oscilloscope, and $R_{eq.}$ is 25Ω . Substitution of the proper values into this equation reveals a peak current in the spark of about 4 amperes (A).

In order to verify the magnitude of the spark current computed in the previous paragraph, a qualitative approxi-

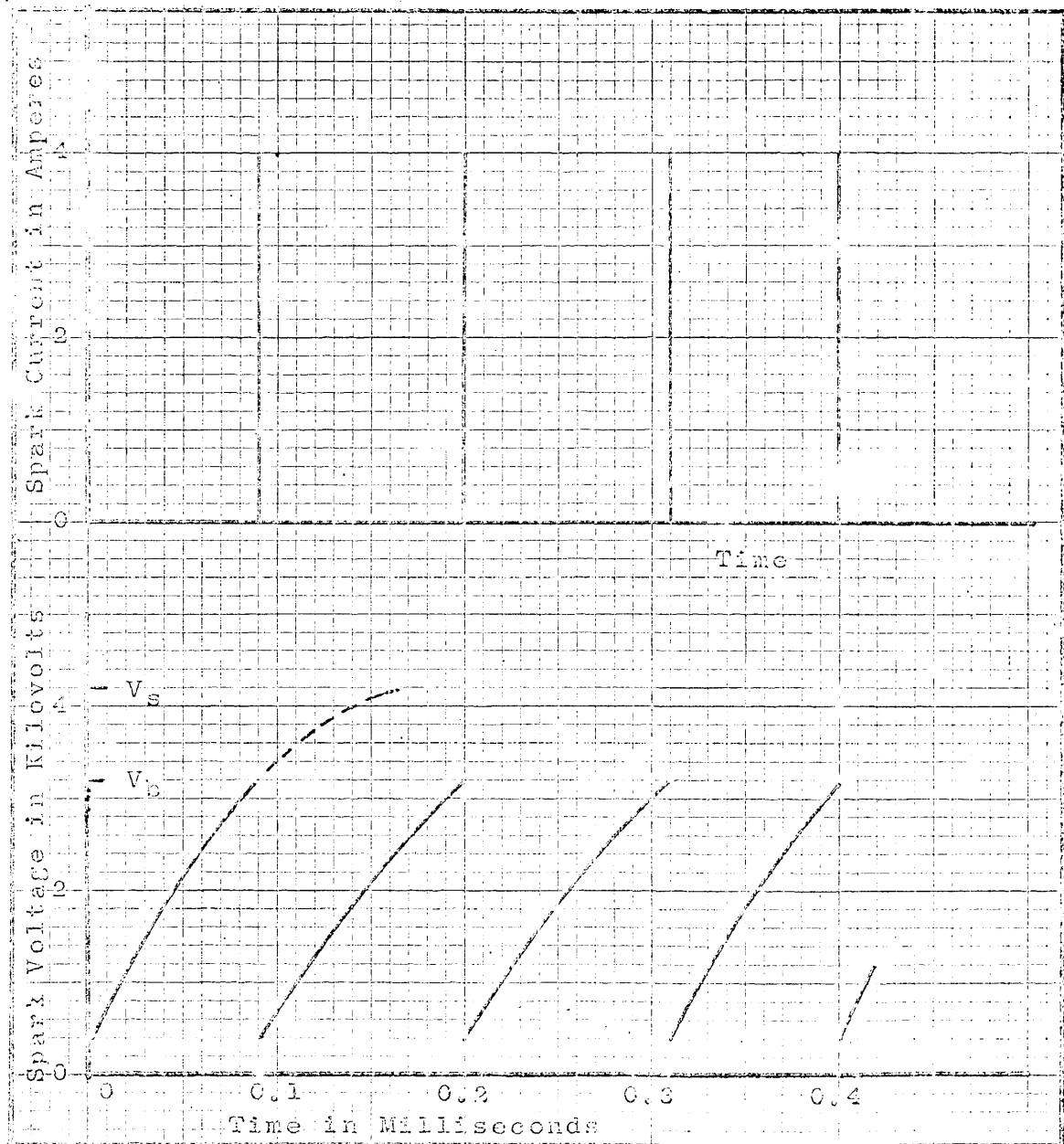


Fig. 3-6 Coincident Plots of Current Pulses vs. Time and Electrode Voltage vs Time. The Current Pulses Represented by Impulses in the Upper Figure Have the Waveform Shown in Figure 3-5.

mation was made of the waveform of the voltage across the electrodes. This voltage was V_s as the spark began and decreased to a minimum of a few hundred volts with a maximum rate of change occurring coincident with i_p . The lower curve of Figure 3-7 shows an estimate of the shape of this waveform with a maximum slope ($\frac{dv}{dt}$) of about 2×10^{12} volts per second. If one assumes an associated capacitance, c , of about 2 picofarads (pF), a peak current is calculated

$$i_p = c \frac{dv}{dt} (\text{max.})$$

to be about 4 A, the same as the measured value.

The Spark Excited by an ac Source

The dc source was replaced with a sinusoidal 60 Hz voltage having a peak value of 6 kV and the spark current and source voltage waveforms observed simultaneously on a dual trace oscilloscope. One half of a cycle of the exciting voltage and the accompanying spark and arc current is shown in Figure 3-8. One observes the spark current begins to exist at point a, the period becomes shorter as the voltage increases and the spark changes to an arc at point b. The arc exists until the source voltage reaches that of point c where the arc changes to a spark and continues until point d is reached. A similar cycle of events occur during the negative half cycle with



Fig. 3-7 Coincident Plots of Spark Current vs. Time and Electrode Voltage vs Time for a 0.030 inch Gap.

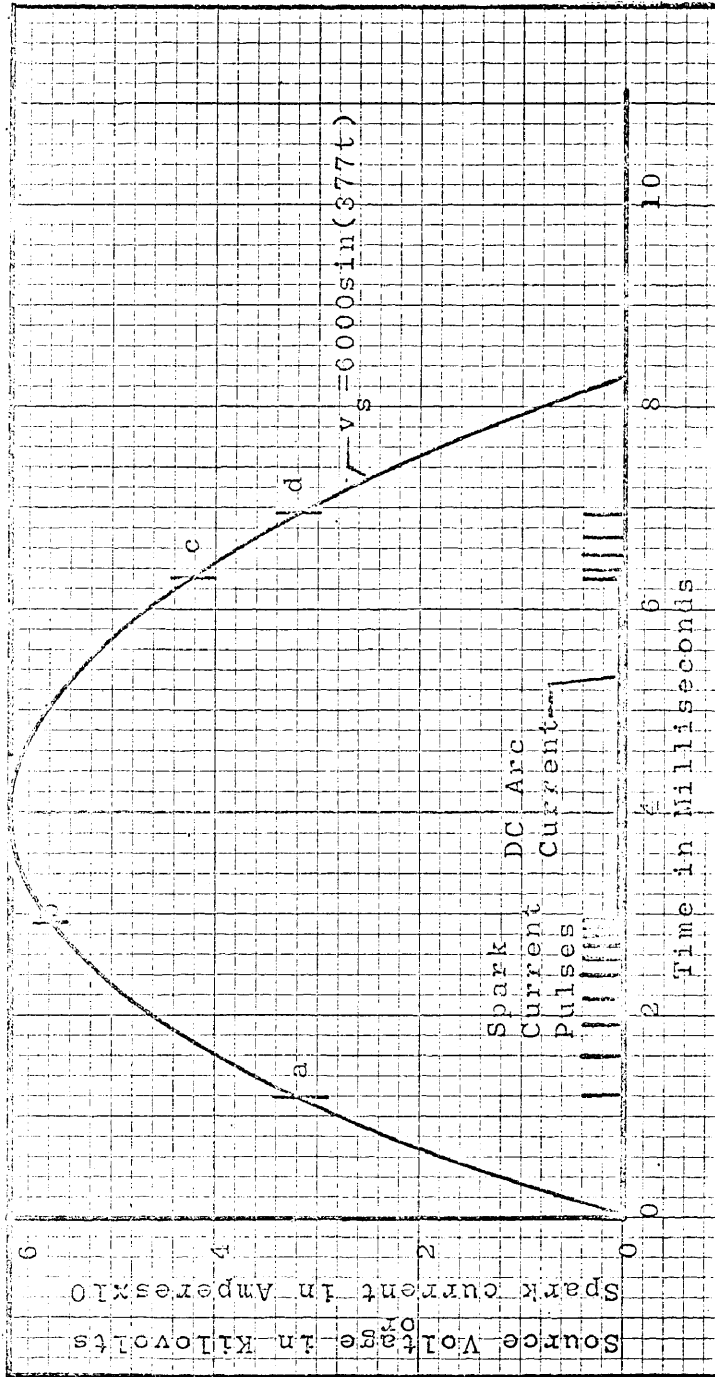


Fig. 3-8 Plot of Spark Current or Arc Current and Source Voltage vs. Time.

voltage and current polarities reversed.

Results of the Time Domain Measurements

The results of the measurements described in this chapter are summarized below in preparation for making the frequency domain measurements and calculations in Chapter IV.

1. The spark characteristics are independent of the type electrode material.
2. The spark amplitude and wave shape is independent of the source voltage for a 0.030 inch (0.762 mm) gap as long as the spark is repetitive.
3. The small current continuous arc has no measurable conduction current at RF (Chapter II).
4. The period of the spark decreases as the source voltage increases.

CHAPTER IV

FREQUENCY DOMAIN CHARACTERISTICS OF THE SPARK

In this chapter the repetitive spark is considered as a broadband noise source with properties which are functions of frequency. A description of the measuring equipment is given, together with the units used to measure the frequency domain characteristics of the noise.

The time domain properties of waveform, amplitude, and period are analyzed in order to determine which properties are essential and which are negligible in this study. The results of the computation of the amplitude versus frequency spectrum are then shown by plots and compared to measured values.

A mathematical expression is given which describes the broadband noise as a function of frequency. This mathematical expression is utilized in the next chapter to calculate radiated field strength when the spark is associated with a radiating element.

The Broadband Signal

The impulsive nature of a periodic spark produces a wide band of frequencies composed of discrete, single-frequency sinusoids. If the spark waveform and period are known, the

amplitude, frequency, and phase of each sinusoidal component may be calculated using Fourier series analysis techniques. If the period is not known or is a random variable, the characteristics of the component sinusoids may not be obtainable in this manner. A description of the method and units used in measuring the broadband noise signal will reveal which of the above named spark parameters are significant and which parameters may be disregarded.

Broadband Noise Measurements

If a broadband noise voltage is applied to the input of a narrow band amplifier whose output is connected to a peak-reading detector, the detector voltage will be a function of the amplifier bandwidth [11]. If the noise is composed of discrete sinusoids not harmonically related, one may assume that occasionally the peaks of all sinusoids exist at the same time. At this time one may write an expression for the voltage V_{peak} of the peak reading detector

$$V_{\text{peak}} = \sum_{n=1}^k A_n \text{ volts} \quad (\text{IV-1})$$

where

k = number of sinusoids supplied by the bandlimited
amplifier to the detector

A_n = amplitude of the n th sinusoid in volts.

The derivation of equation (IV-1) applies also to complex wave forms composed of harmonically related sinusoids if their phases are such that all of the peaks of the sinusoids concur periodically. In a similar manner, Schwartz [12] shows that a noise whose voltage amplitude spectrum is continuous will produce a peak voltage whose value is

$$V_{\text{peak}} \approx \int_{f_2}^{f_1} A(f) df \quad (\text{IV-2})$$

where

f_1 = lower band limit of the amplifier in Hz

f_2 = upper band limit of the amplifier in Hz

$A(f)$ = the voltage amplitude spectral density in volts per Hz.

If the amplifier bandwidth is narrow compared to the bandwidth of the broadband signal and the amplitudes of the components of the broadband signal do not change rapidly with frequency, the component amplitudes will be approximately constant in the pass-band. Then the peak reading detector output voltage will be directly proportional to the amplifier bandwidth and may be computed as follows:

$$V_{\text{peak}} \approx AN \text{ volts per bandwidth} \quad (\text{IV-3})$$

where

A = average value of the amplitude of the sinusoids
in the passband

N = number of components in the passband.

Thus, the broadband noise voltage may be measured and described in terms of peak voltage per unit bandwidth [13].

The units used in this report were microvolts (μV) per MHz, decibels (dB) above one μV per MHz, or amperes (A) per MHz when describing direct measurements on the spark, or dB above one μV per meter (m) per MHz when describing radiated field strengths.

Measurement Equipment

The frequency domain characteristics of the noise were measured with a Fairchild Electro-Metrics Corporation model EMC-25 Interference Analyzer. This unit is a frequency selective RF voltmeter calibrated to indicate the root-mean-square (rms) value of a sinusoidal voltage at the input terminals. When a non-sinusoidal waveform is measured, with the detector in peak mode, the meter reading is in terms of an equivalent sinusoidal voltage whose peak value is the same as that of the non-sinusoidal waveform being measured. The meter reading units are μV rms or dB above one μV rms. In addition, when one is measuring a broadband noise voltage, one compensates for the analyzer bandwidth by using a

correction factor supplied by the manufacturer. For example, if the bandwidth of the analyzer is 5 kHz, its reading is converted to μV per MHz by multiplying the analyzer reading by $200 \{10^6 \text{ divided by } (5)(10^3)\}$.

A calibrated antenna is connected to the EMC-25 when one measures the field strength of a broadband noise radiation. This requires an additional antenna efficiency correction factor (supplied by the manufacturer) to convert the voltage from the calibrated antenna to a field strength reading in μV per m per MHz. The measurement apparatus schematic diagram is shown in Figure 4-1.

The Effect of the Pulse Period

A periodic function of time $f(t)$ with period T may be represented by an infinite series of sine and cosine terms, called a Fourier series, as follows:

$$f(t) = a_0 + \sum_{n=1}^{\infty} \{a_n \cos(n\omega_1 t) + b_n \sin(n\omega_1 t)\} \quad (\text{IV-4})$$

where

a_0 = average value of $f(t)$

a_n = amplitude of the n th cosine term

b_n = amplitude of the n th sine term

n = number of the harmonic frequency

ω_1 = fundamental frequency in radians/second.

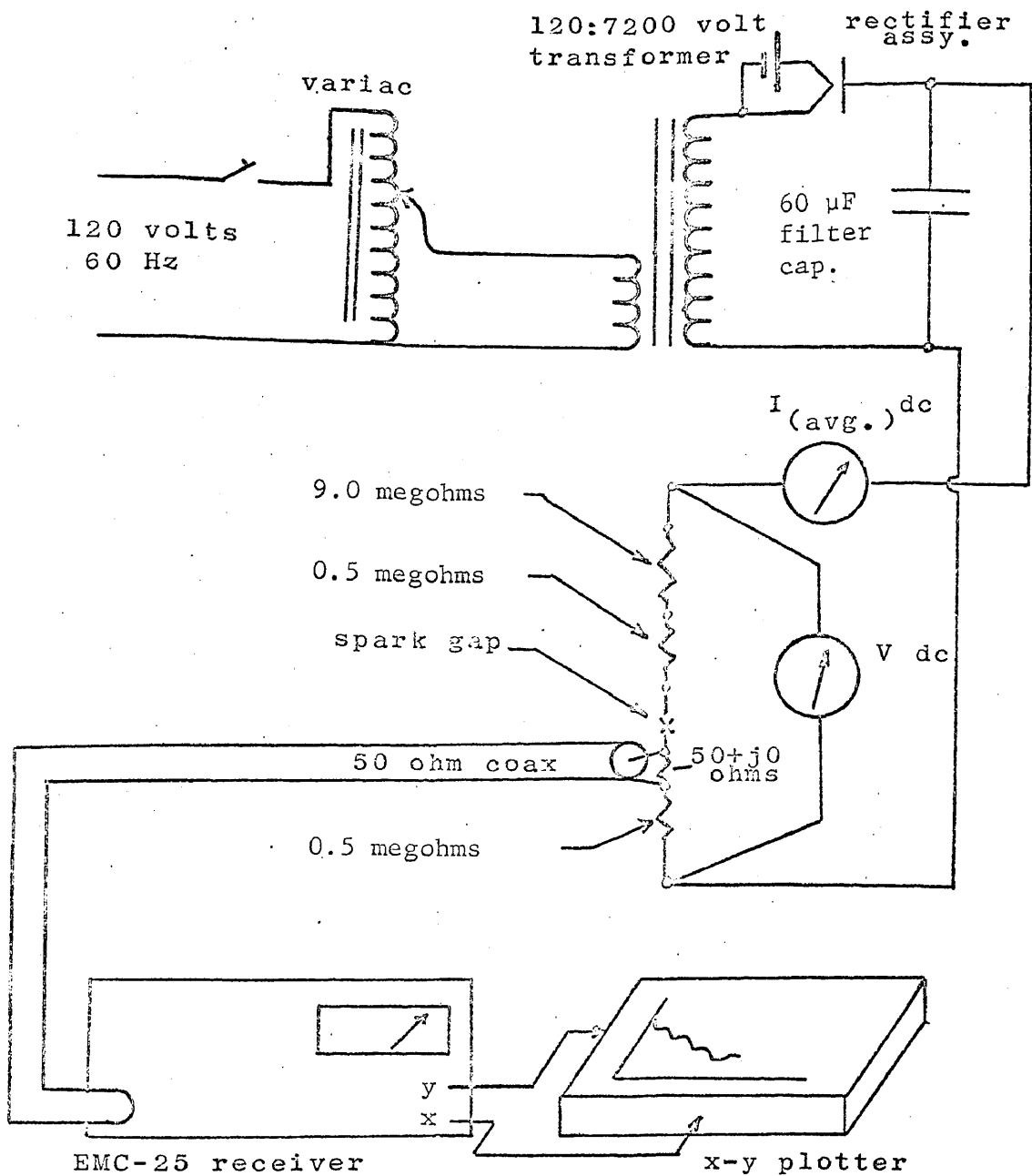


Fig. 4-1 Schematic Diagram of Frequency Domain Measurement Apparatus.

One computes the value of ω_1 by

$$\omega_1 = \frac{2\pi}{T}. \quad (\text{IV-5})$$

The coefficients of the series terms are calculated as follows:

$$a_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) dt \quad (\text{IV-5a})$$

$$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \cos(n\omega_1 t) dt \quad (\text{IV-5b})$$

$$b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \sin(n\omega_1 t) dt. \quad (\text{IV-5c})$$

The average value a_0 is the dc component of the function and was not of interest in this study. If the leading and trailing edges of the pulse waveform are similar with slopes which are equal in magnitude but the negative of each other in sign, one may choose the origin such that $f(t)$ is an even function. The Fourier series representing an even periodic function is a cosine series [14] and the b_n are zero for all n . The function

$f(t)$ may then be written

$$f(t) = \sum_{n=1}^{\infty} a_n \cos(n\omega_1 t). \quad (\text{IV-6})$$

If this waveform is measured as described in the section titled Broadband Noise Measurements in this dissertation the peak detector reading may be computed as follows:

$$V_{\text{peak}} = A_c N_{\beta} \text{ volts per bandwidth} \quad (\text{IV-7})$$

where A_c is the amplitude in volts of the component whose frequency coincides with the center of the analyzer passband and N_{β} is the number of components in the passband. One obtains the value of N_{β} from the relationship

$$N_{\beta} = \frac{2\pi\beta}{\omega_1} = \frac{\beta}{f_1} = \beta T \quad (\text{IV-8})$$

where β is the analyzer bandwidth in Hz and f_1 is the value of the lowest frequency component in Hz. If one substitutes equations (IV-8) and (IV-5b) into equation (IV-7) one obtains

$$V_{\text{peak}} = (\beta T) \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \cos(N_c \omega_1 t) dt \quad (\text{IV-9})$$

where N_c is the number of the passband center frequency component.

The value of N_c may be obtained from the relationship

$$N_c = \frac{\omega_c}{2\pi f_1} = \frac{\omega_c T}{2\pi} \quad (\text{IV-10})$$

where ω_c is the radian center frequency. One may combine equations (IV-10) and (IV-5) to show that the argument of the cosinusoid in equation (IV-9) is

$$N_c \omega_1 t = \left(\frac{\omega_c T}{2\pi} \right) \left(\frac{2\pi}{T} \right) t = \omega_c t \quad (\text{IV-11})$$

and is thus independent of T .

The spark waveform has a pulse width b much shorter than its period and its amplitude is approximately zero between pulses i.e. $\frac{b}{2} < t < (T - \frac{b}{2})$. The limits of integration of equation (IV-9) may therefore be changed to $-\frac{b}{2}$ and $\frac{b}{2}$ and equation (IV-9) becomes:

$$V_{\text{peak}} = 2\beta \int_{-\frac{b}{2}}^{\frac{b}{2}} f(t) \cos(\omega_c t) dt \quad (\text{IV-12})$$

This equation shows that the reading of the peak detector, when measuring a broadband noise at frequency ω_c , is a

function of the bandwidth β of the amplifier as well as the waveform and amplitude of the spark current but is independent of the period T .

Equation (IV-12), which is independent of the spark period, was derived with the assumption that the spark waveform was symmetric about $t = 0$ such that the waveform was an even function of time. This assumption was made for simplification purposes and may be relaxed to permit any arbitrary waveform whose pulse width is narrow compared to its period. If the waveform is not an even function, equation (IV-12) becomes [15]:

$$V_{\text{peak}} = 2\beta \left[\left\{ \int_{-\frac{b}{2}}^{\frac{b}{2}} f(t) \cos(\omega_c t) dt \right\}^2 + \left\{ \int_{-\frac{b}{2}}^{\frac{b}{2}} f(t) \sin(\omega_c t) dt \right\}^2 \right]^{\frac{1}{2}} \quad (\text{IV-13})$$

Laboratory measurements performed during this study verified that the amplitude of the noise voltage as measured by the EMC-25, operating in peak detector mode, was independent of the period. The dc source voltage was changed from minimum sparking potential to six kV. This caused a large variation in the period but produced no change in the measured value

of the noise voltage. Similarly, when the dc source voltage was replaced with a 60 Hz sinusoidal voltage high enough so that a repetitive spark existed over part of the cycle, the measured value of the noise voltage was the same as was obtained with the dc source.

Since the amplitude of the noise voltage was independent of the spark period, calculations were made using an arbitrary period of 10^{-4} second.

The Frequency Domain Spectrum

The calculated amplitude of the noise voltage in volts per unit bandwidth was obtained by performing numerical integration using the trapezoidal rule. A suitable program, given in Appendix A, was written for the IBM 360-65 digital computer using time intervals of 0.1 nanosecond (ns). The width of the main part of the spark pulse was about two ns but the integration interval b was about six ns to include all of the time domain waveform as shown in Figure 3-5.

The results of the calculations described above are shown in the graph of Figure 4-2 in which the units of the ordinate are dB above 1 μ V/MHz and the units of the abscissa are MHz. One observes that the amplitude spectrum is flat to about 300 MHz and then begins to decrease but is still significant at one GHz. Even at two GHz (well beyond the capabilities of

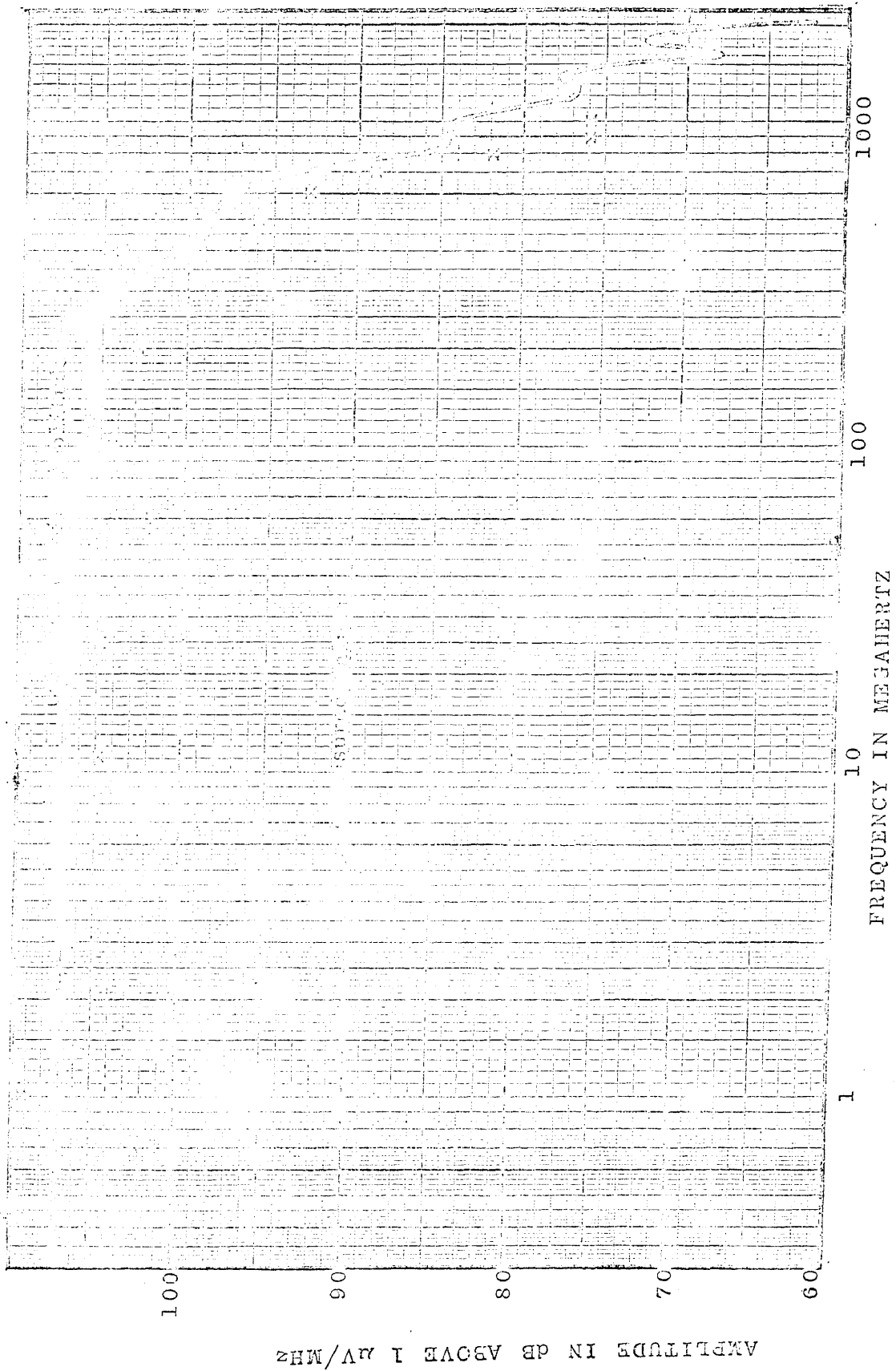


Fig. 4-2 Plot of Noise Voltage Amplitude Calculated from Time Domain Waveform.

the measuring equipment) the calculated amplitude is still 65 dB above one $\mu\text{V}/\text{MHz}$.

Measured values of the noise voltage are shown at discrete frequencies in Figure 4-1 for comparison purposes. The measured values differ by no more than 4 dB from the calculated values for frequencies between one and 800 GHz.

It is estimated that the EMC-25 Interference Detector is accurate to within about ± 4 dB when measuring broadband voltages. This estimate is based on a comparison of measurements made on the same noise voltage at the same frequency but on opposite ends of adjacent bands. An example of a difference in measured values was obtained from the measurements made at 500 MHz as outlined above. At the high frequency end of band 14 (200 MHz to 500 MHz) the noise voltage measured 100 dB above 1 μV and at the low frequency end of band 15 (500 MHz to 1000 MHz), the same noise voltage measured 96 dB above 1 μV . The calculated value, 98 dB above 1 μV , is between the two measured values. Samples of measured values are shown in Appendix B.

The Noise Voltage Expressed Mathematically

An analytic expression for the noise voltage in volts per unit bandwidth was obtained by using the log-magnitude versus log-frequency curve of Figure 4-2 to construct a "Bodé diagram" type plot. The procedure used to produce an expression was

based on a generalized form as a ratio of polynomials [16,17]

as follows:

$$F(j\omega) = \frac{K(j\omega)^N \prod_{i=1}^Q (1 + j\omega\tau_i)}{(j\omega)^L \prod_{m=1}^M (1 + j\omega\tau_m) \prod_{k=1}^R \{1 + A_k\tau_k(j\omega) + (j\omega\tau_k)^2\}} \quad (\text{IV-14})$$

where

$F(j\omega)$ = a function of frequency

K = a constant

L, N = integers

$\frac{1}{\tau_m}, \frac{1}{\tau_k}, \frac{1}{\tau_i}$ = break frequencies in radians per second.

The values assigned to K , L , N , and the break frequencies are determined by the particular function being represented by expression (IV-14).

The special form of expression (IV-14) which is useful in representing mathematically the curve of Figure 4-2 is obtained by setting L , N , τ_i , and τ_k equal to zero and rewriting equation (IV-14) as follows:

$$F(j\omega) = \frac{K}{\prod_{m=1}^M (1 + j\omega\tau_m)} \quad (\text{IV-15})$$

The function $F(j\omega)$ is a complex quantity and its logarithm is also complex as may be shown by the following relationship [18]:

$$\ln F(j\omega) = \ln(|F(j\omega)| e^{j\phi(\omega)}) \quad (\text{IV-16a})$$

$$= \ln |F(j\omega)| + \ln e^{j\phi(\omega)} \quad (\text{IV-16b})$$

$$= \ln |F(j\omega) + j\phi(\omega)| \quad (\text{IV-16c})$$

The logarithm of the magnitude of a function of frequency $F(j\omega)$ expressed in decibels (dB) is

$$\text{Lm}F(j\omega) = 20 \log_{10} |F(j\omega)| \text{ dB} \quad (\text{IV-17})$$

and is called the log-magnitude and is abbreviated $\text{Lm}F(j\omega)$ [19].

Since $|F(j\omega)|$ is a function of frequency, $\text{Lm}F(j\omega)$ is also a function of frequency.

The expression for $\text{Lm}F(j\omega)$ may be expanded as follows:

$$\begin{aligned} \text{Lm}F(j\omega) &= \text{Lm} \left[\frac{K}{(1 + j\omega\tau_1)(1 + j\omega\tau_2) \dots (1 + j\omega\tau_m)} \right] \\ &= \text{Lm} K - \text{Lm}(1 + j\omega\tau_1) - \text{Lm}(1 + j\omega\tau_2) \\ &\quad - \dots - \text{Lm}(1 + j\omega\tau_m). \end{aligned} \quad (\text{IV-18})$$

The plot of $\text{Lm}F(j\omega)$ versus log-frequency will approach the

value of $L_m K$ for frequencies well below the smallest $\frac{1}{\tau_m}$.
 for frequencies well above the largest $\frac{1}{\tau_m}$, the slope
 of the curve will approach -20 dB per decade increase
 in frequency for each term containing a τ_m .

The function of radian frequency $F(j\omega)$ may be changed
 to a function of frequency in Hz by making the substitution

$$\omega = 2\pi f \quad (\text{IV-19})$$

where f is Hz, allowing one to rewrite equation (IV-15) as
 follows:

$$\begin{aligned} F(f) &= \frac{K}{\prod_{m=1}^M \left(1 + \frac{j2\pi f}{2\pi f_m}\right)} \\ &= \frac{K}{\prod_{m=1}^M \left(1 + j \frac{f}{f_m}\right)} \end{aligned} \quad (\text{IV-20})$$

where

$F(f)$ = a function of frequency in Hz

f_m = a break frequency in Hz

f = the independent variable in Hz.

Referring to Figure 4-1, one observes the amplitude of the
 noise voltage was constant at 107 dB above 1 μ V/MHz for frequen-
 cies below 10 MHz, permitting one to choose the low frequen-

cy asymptote parallel to the abscissa and equal to 107 dB above 1 $\mu\text{V}/\text{MHz}$. The amplitude of the noise voltage was observed to decrease at a rate approximately proportional to the inverse of the frequency cubed for frequencies above 500 MHz, permitting a choice of a high frequency asymptote with a slope of -60 dB per decade. If one chooses f_1 , f_2 , and f_3 equal to 500 MHz, the high frequency asymptote will intersect the low frequency asymptote at 500 MHz, as shown in Figure 4-3. Application of the required corrective factors [20] to the asymptotic curve resulted in the final curve marked $V(f)$ also shown on Figure 4-2 which is a plot of L_m versus log-frequency of the equation

$$\begin{aligned}
 V(f) &= \left| \frac{(2)(10^5)}{\left\{1 + j \frac{f}{(0.5)(10^9)}\right\} \left\{1 + j \frac{f}{(0.5)(10^9)}\right\} \left\{1 + j \frac{f}{(0.5)(10^9)}\right\}} \right| \\
 &= \left| \frac{(2)(10^5)}{\{1 + j(2)(10^{-9})(f)\}^3} \right| \\
 &= \frac{(2)(10^5)}{\{1 + (4)(10^{-18})(f^2)\}^{\frac{3}{2}}} \quad \mu\text{V}/\text{MHz}. \quad (\text{IV-21})
 \end{aligned}$$

Recall that the voltage which is approximated by equation (IV-21) was produced by the spark current existing in a 50 ohm resistor connected to a 50 ohm co-axial cable. The spark

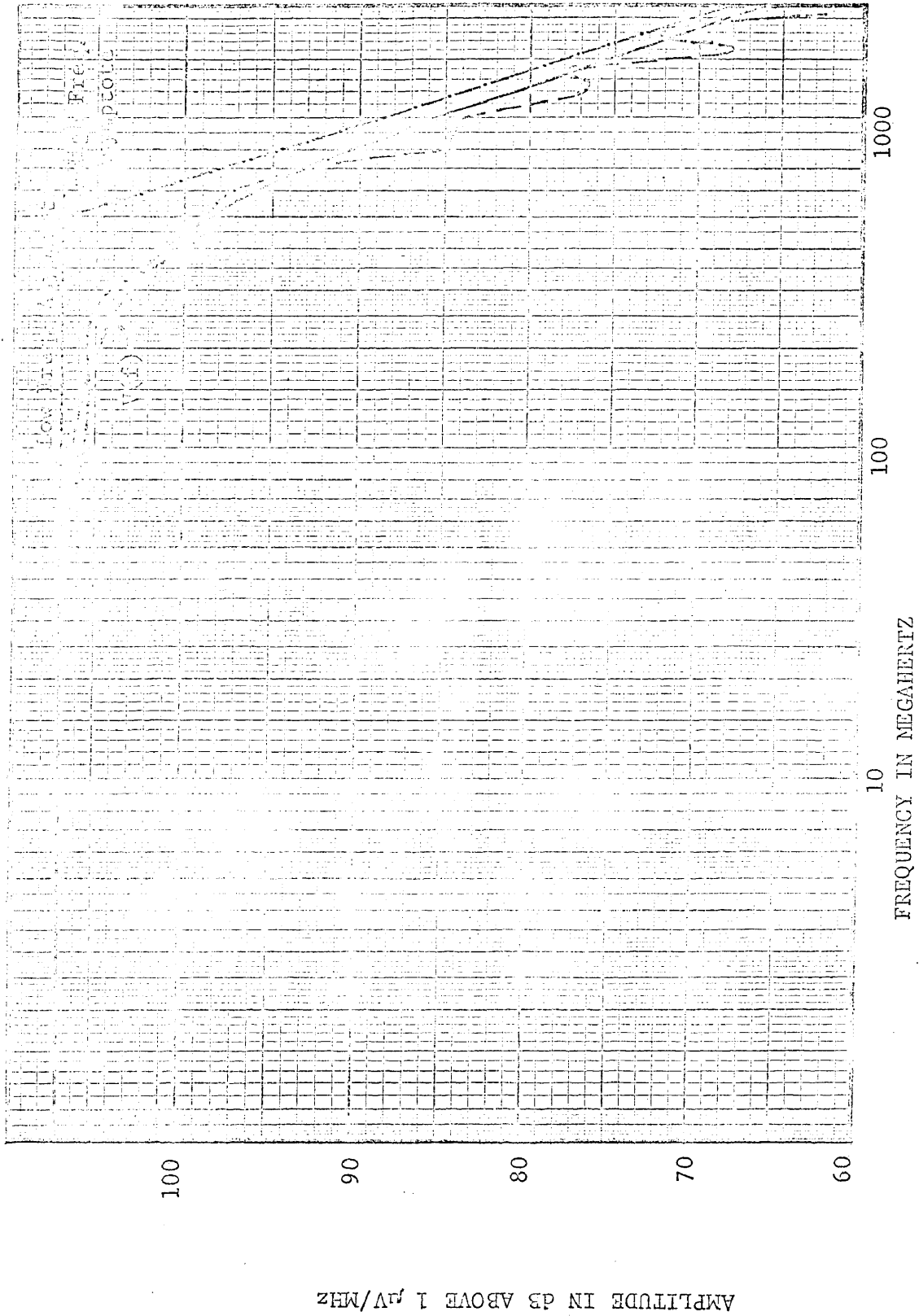


Fig. 4-3 Plot of $V(f)$ Calculated from Equation (IV-20)

current was therefore calculated as follows:

$$I(f) = \frac{V(f)}{R_{eq}} \quad \mu\text{A per MHz} \quad (\text{IV-22})$$

where $R_{eq} = 25$ ohms.

An equivalent circuit to represent the spark as a two-terminal model was chosen to be a current source because current was the measured characteristic of the spark. Realizable current sources must be associated with a shunt admittance, resulting in the following model:

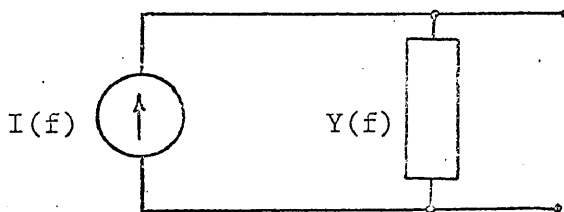


Fig. 4-4 Preliminary Model of the Spark.

where $Y(f)$ is a complex admittance which may be a function of frequency.

The high source voltage which existed across the spark gap plus the low input impedance (50 ohms) of the EMC-25 prevented direct voltage measurements on the spark. The magnitude of $Y(f)$ was therefore not obtainable from measurements.

Values for the circuit elements of the parallel admittance $Y(f)$ were assumed as outlined below. Subsequent measurements showed the assumed values agreed with measured values. A ten megohm resistor R_s was in series with the spark gap and the source voltage, so $Y(f)$ was assumed to have a conductance of 10^{-7} mho. Although the conductance of the air between electrodes increased for a few ns while the ionized spark path existed, during most of the period the air was non-conducting and its effect on the shunt conductance was neglected. The distributed capacitance associated with a spark gap on an electric power line was approximated by a 4 picofarad (pF) capacitor placed in parallel with the electrodes. The susceptive part of $Y(f)$ was therefore assumed to be capacitive and the equivalent circuit model was as follows:

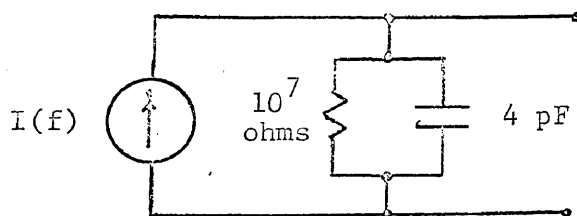


Fig. 4-5 Final Model of the Spark.

The model portrayed in Figure 4-5 was assumed to represent the spark noise as an ideal current source shunted by an admittance.

Summary of Chapter IV

In this chapter, a description has been given of the frequency domain characteristics of the spark. The measuring equipment has been described and the units used in this study were given. It was shown that the frequency domain measurements are independent of the period of the spark. Finally, a mathematical expression was derived and a model was developed to describe the noise source.

The next chapter describes the measurements of the radiated electromagnetic field produced by a spark between a pair of electrodes representing pole hardware items. The spark is represented by the model shown in Figure 4-4 and the hardware items are represented by a dipole antenna driven at its center.

CHAPTER V

THE ELECTROMAGNETIC RADIATION

PRODUCED BY THE SPARK

This chapter contains a report on the measured and calculated values of the electromagnetic radiation produced by a spark noise source driving a dipole antenna. The reasons for the choice of the dipole radiator are given and the apparatus used to produce the noise source is described. The results of the measurements and calculations are shown by plots and some qualitative information about the impedance of the spark is obtained.

The Radiating System

The sparks which exist on electric power lines may occur between various items of pole line hardware. For example, a spark may form between a high voltage conductor and an improperly installed tie wire, in which case the primary radiating element is the conductor acting as a long-wire traveling-wave antenna.

If the spark exists between a ground wire and a cross-arm mounting bolt, the ground wire may act like a vertical monopole radiator. A third configuration might result from the spark forming at the junction of a pin and clevis in a string of bell-type insulators, or at the center mounting

bolt of a pair of galvanized steel cross arm braces. The third configuration described produces a radiating system which is basically a center driven dipole.

Each of the configurations described above produces a primary radiating system. It is recognized, of course, that capacitive and inductive coupling exists between many components on electric power lines, so that there may be multiple secondary radiators associated with the primary system. However, only the primary radiating system was considered in this study.

The choice of the radiating system investigated in this study was influenced by several factors, including the relative frequency of their occurrence and the availability of an analytical description of the radiated field of the system.

On power lines, many sparks are found across air gaps between items of hardware which form a primary radiating system of a center driven dipole. In addition, the relationship between the current in a dipole and the resulting distant radiation field is well-known [21]. Both of these factors led to the choice of a dipole radiator to be associated with the spark noise source in this investigation.

The Dipole Radiator

The dipole radiators representing typical radiating

elements on power lines were chosen with lengths as shown below in Table (V-1), with corresponding resonant frequencies.

TABLE (V-1)

List of Dipole Lengths, Center Capacitance,
and Resonant Frequencies

Dipole Length in Meters	Center Cap. in pF	Half-wave Resonant frequency in MHz	Full wave Resonant frequency in MHz
0.732	6	200	400
0.357	5	400	800
0.237	4	600	1200
0.177	3	800	1600
0.157	2	900	1800

Note that one column lists the resonant frequencies corresponding to the dipole being driven as a half-wave antenna and another column lists the resonant frequencies corresponding to the dipole being driven as a full wave antenna. The lengths, as half-wave resonant dipoles, were calculated from the relationship given by Kraus [22] as follows:

$$L = \frac{0.48 \lambda (L/D)}{L/D + 1}$$

$$= 0.48 \lambda - D \text{ meters} \quad (V-1)$$

where

L = antenna length in meters

D = antenna diameter in meters

λ = wave length in meters.

All elements were made from 0.125 inch (0.00313 m) brass stock.

The measured capacitance at the center of the dipoles is included in Table (V-1) also. This capacitance, because of the high frequencies involved, cannot be viewed as a lumped parameter, but should be considered as a distributed capacitance. However, the direct measurements on the spark were made with this distributed capacitance represented by a lumped capacitance of two to four picofarads (pF).

The Measured Radiated Field

The distant radiated electric field strength measurements were made using an arrangement as shown in Figure 5-1. The spark noise source was produced by causing a spark to exist across a 0.030 inch (0.762 mm) air gap between steel electrodes identical to those used when making the direct measurements described in Chapter IV. The spark was excited by a 4200 V rms sixty Hz source connected to the electrodes through two 5 megohm resistors. The dipole elements were then attached to the steel electrodes by a plug and socket arrangement, producing a symmetrical, balanced, center driven radiating system. The dipoles were mounted horizontally on a polystyrene

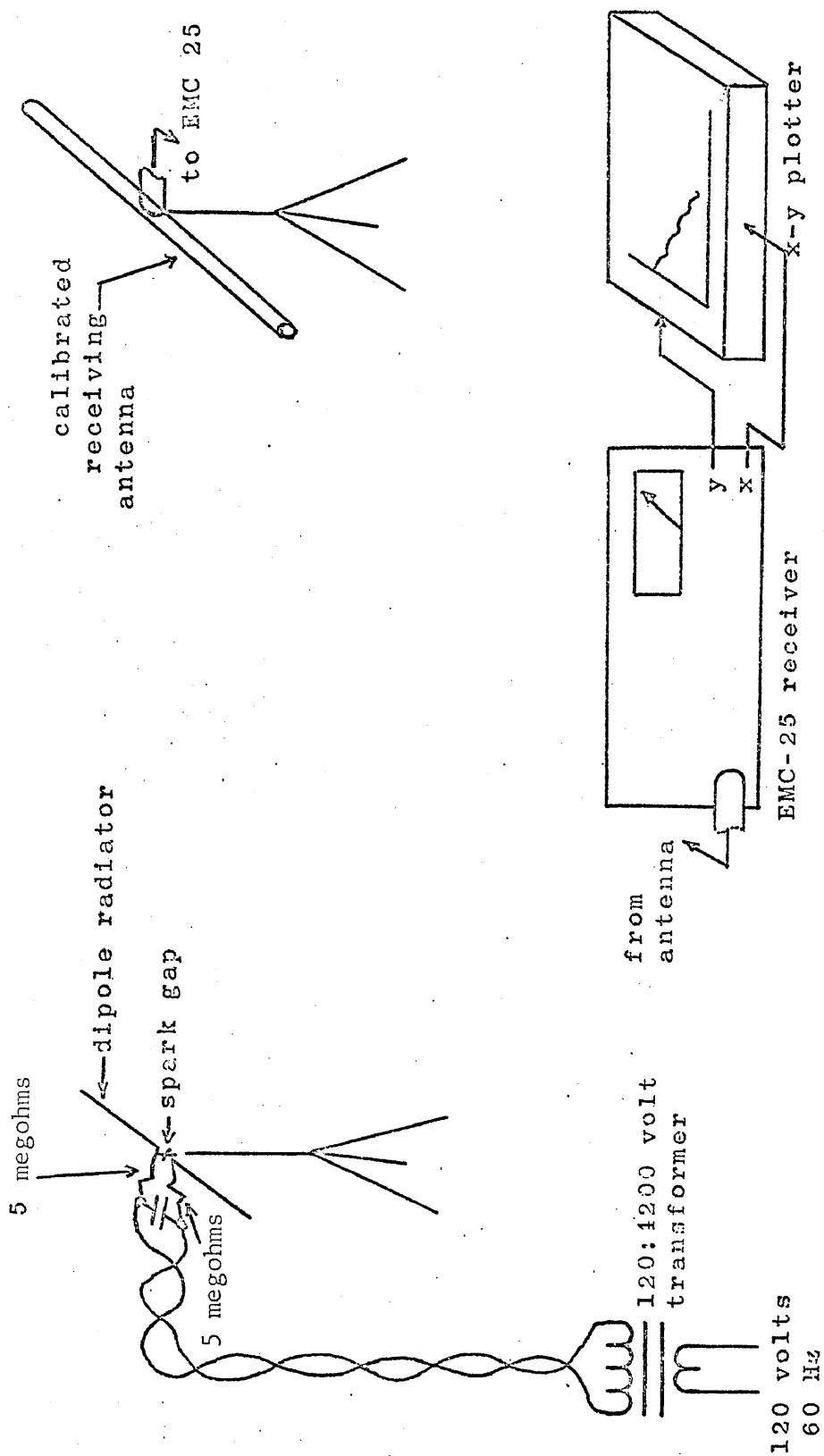


Fig. 5-1 Schematic Diagram of Apparatus Used to Measure Radiated Field Strength.

mast which elevated the dipoles ten feet (3.048 m) above the roof of Bolton Hall.

The radiated electric field strength was measured with the EMC-25 Noise Detector in conjunction with calibrated antennas as described in Chapter IV. Measurements at frequencies from 50 to 200 MHz were made using three lengths of calibrated dipoles mounted horizontally, six feet (1.83 m) above the roof on a wooden tripod. Measurements made at frequencies from 200 to 1000 Mhz were made using a circularly polarized log-periodic-conical antenna, also mounted six feet (1.83 m) above the roof.

The effect of nearby absorbing and reflecting materials was minimized by making the measurements with all equipment mounted on the roof of a 44 foot (13.4 m) high masonry building. The transmitting and recieving antennas were placed on the corners of an "L" shaped part of the building so that most of the distance between the antennas was free space 44 feet (13.4 m) above the earth.

The measured values of the radiated electric field are shown in Figure 5-2. Frequency in MHz is the independent variable, field strength in dB above one $\mu\text{V}/\text{m}/\text{MHz}$ is the dependent variable and dipole length is a parameter.

Calculated Radiated Field

The general expression for the magnitude of the radiation

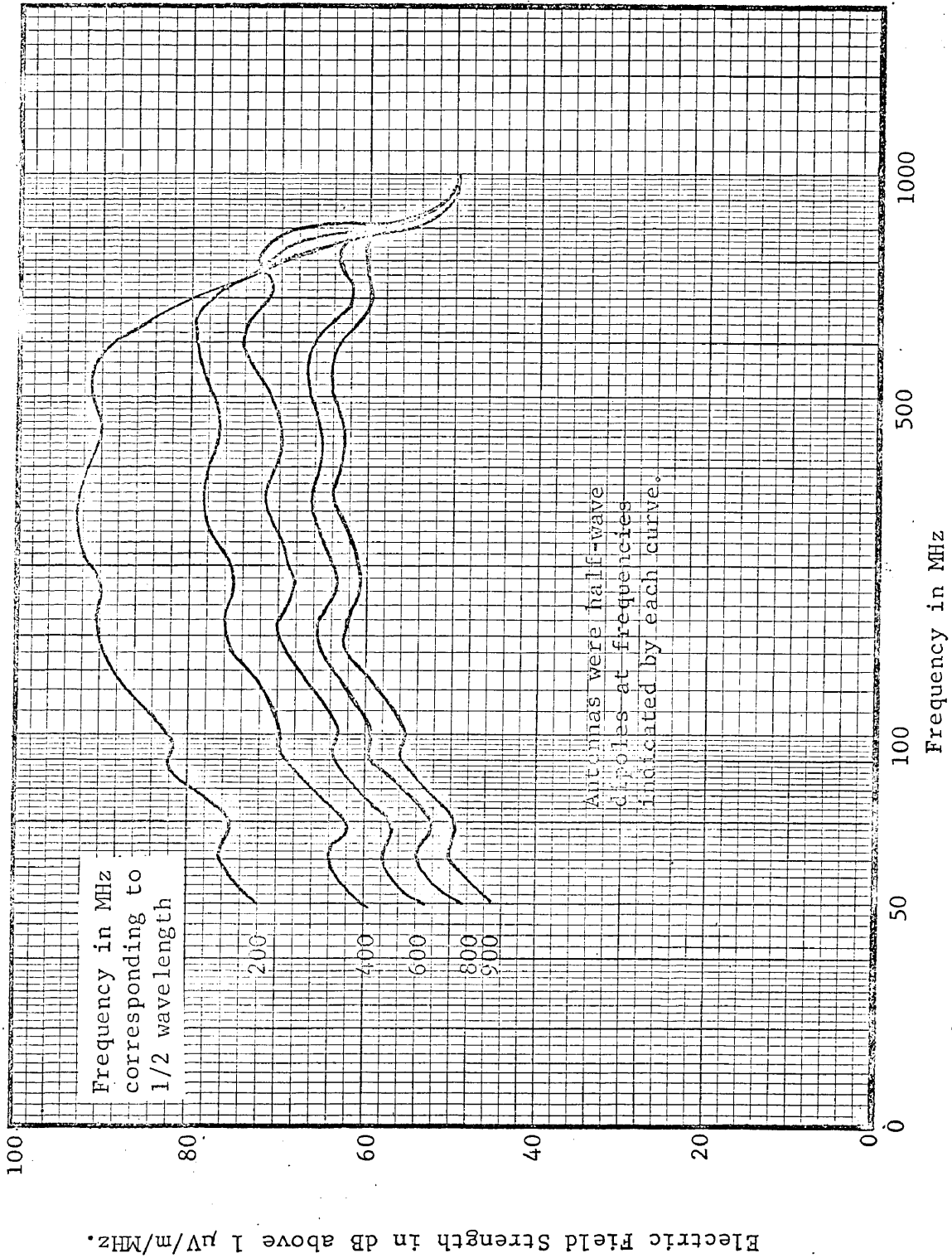


Fig. 5-2 Measured Radiated Electric Field Strength 21.35 Meters from Dipole.

field strength due to a dipole of any length $L = 2H$ is given by Jordan [23] as follows:

$$E = \frac{60 I}{r} \left[\frac{\cos(\beta H) - \cos(\beta H \cos \theta)}{\sin \theta} \right] \text{ volts/meter (V/m)} \quad (\text{V-2})$$

where

E = the magnitude of the electric field intensity in V/m

r = the distance from the center of the dipole in m

θ = the angle in radians between r and a line collinear with the dipole elements

$\beta = 2\pi$ divided by wave length in meters

H = total antenna length in meters divided by two

I = peak value of the current at the current maximum, in amperes.

For antenna lengths up to one wave length, the radiation pattern of a dipole consists of two major lobes normal to the antenna. In this study, the antenna length was limited to one full wave length or less, and all measurements were made perpendicular to the antenna so that θ in equation (V-2) was $\pi/2$ radians and the equation becomes

$$E = \frac{60 I}{r} [\cos(\beta H) - 1] \quad (\text{V/m}) \quad (\text{V-3})$$

The electric field strength E calculated by equation (V-3)

gives the peak value of a sinusoidal function of time. The independent variable in equation (V-3) is a current with a peak value I also varying sinusoidally with time. Furthermore equation (V-3) was derived with an assumed sinusoidal current distribution along the antenna and current nodes at each end and for this study these assumptions are also made.

If the current I in equation (V-3) was composed of several or even many sinusoids, one could consider each component individually and calculate the electric field strength as the sum of the effects of each component. In a similar manner, it will be assumed the independent variable I can become a band of frequencies, i.e., current per unit bandwidth and the electric field strength E can then be measured in terms of volts per meter per unit bandwidth. If $I(f)$ in peak μA per MHz as calculated by equation (IV-21) is inserted in equation (V-3), the result would be the electric field intensity in peak μV per meter per MHz. An expression which combines equations (V-3) and (IV-21) is

$$E(f) = \frac{(8)(10^3)}{\{1 + (4)(10^{-18})(f^2)\}^{\frac{3}{2}}} \frac{60}{r} [\cos(\beta H) - 1] \quad (V-4)$$

The argument of the cosine term may be rewritten

$$\beta H = \frac{2\pi H}{\lambda}$$

$$\begin{aligned}\beta H &= \frac{\pi L}{\lambda} \\ &= (3.33)(10^{-9}) \pi L f\end{aligned}\quad (V-5)$$

where

f = frequency in Hz

L = antenna length in meters

and equation (V-3) becomes

$$E(f) = \frac{(0.48)(10^6) \{ \cos(3.33)(10^{-9})(\pi)(L)(f) - 1 \}}{\{1 + (4)(10^{-18})(f^2)\}^{\frac{3}{2}}} \mu V/m/MHz. \quad (V-6)$$

Plots of the calculated radiated electric field intensity in dB above 1 μV per m per MHz are shown in Figure 5-3. One observes that for all antenna lengths the curves show a maximum in the frequency range from 300 to 600 MHz. For frequencies below 200 MHz, the curves approach a constant slope of 40 dB per decade. The curves should approach a slope of -20 dB per decade for frequencies above 600 MHz, but the curves were not continued past one GHz because of limitations of the measuring equipment.

Comparison of Measured and Calculated Results

A plot of measured values and calculated values of the radiated electric field are shown in Figure 5-4. These values are for the dipole whose half-wave resonant frequency was 400 MHz, but the curve is typical of the other antennas.

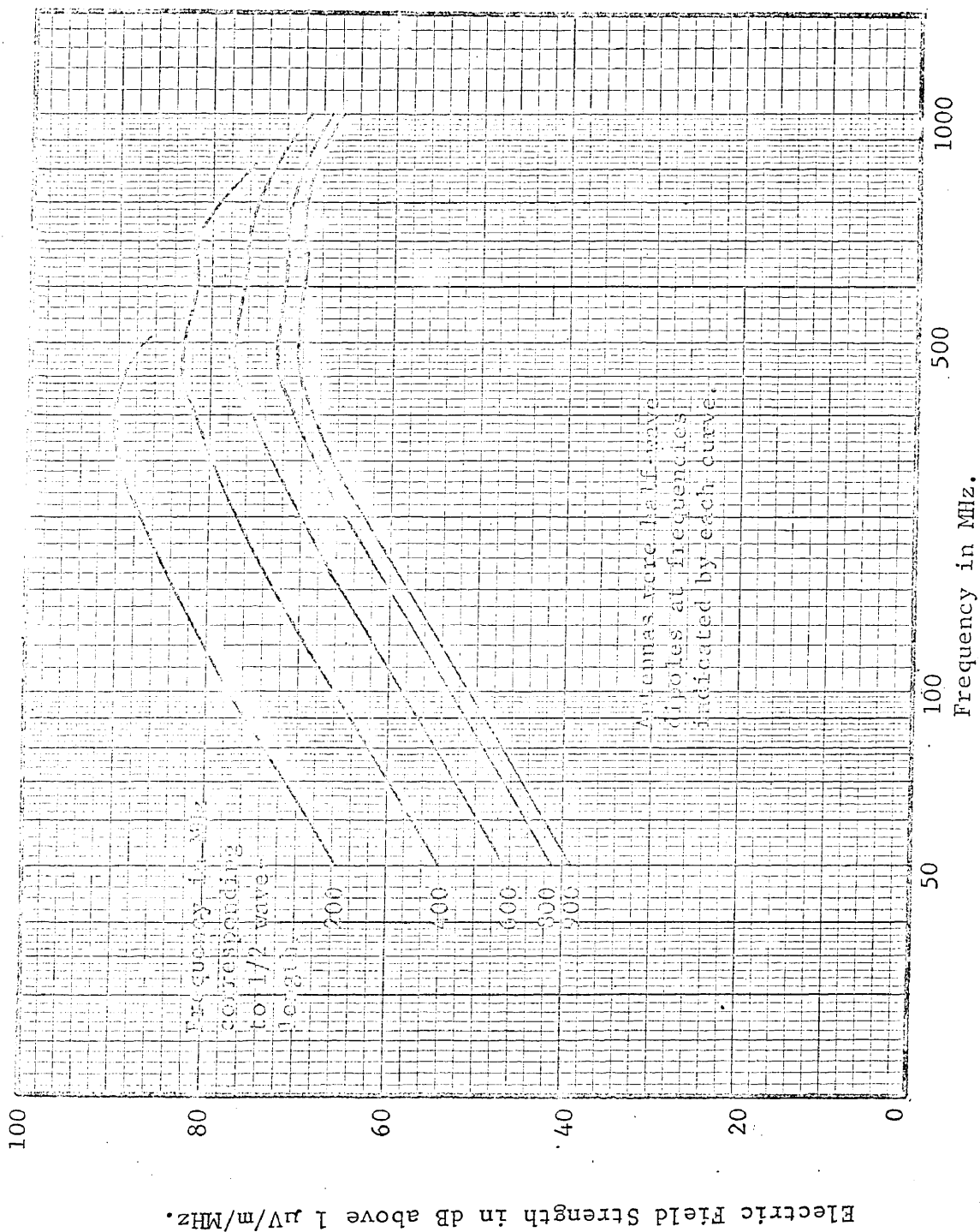


Fig. 5-3 Calculated Radiated Electric Field Strength 21.35 meters from dipole.

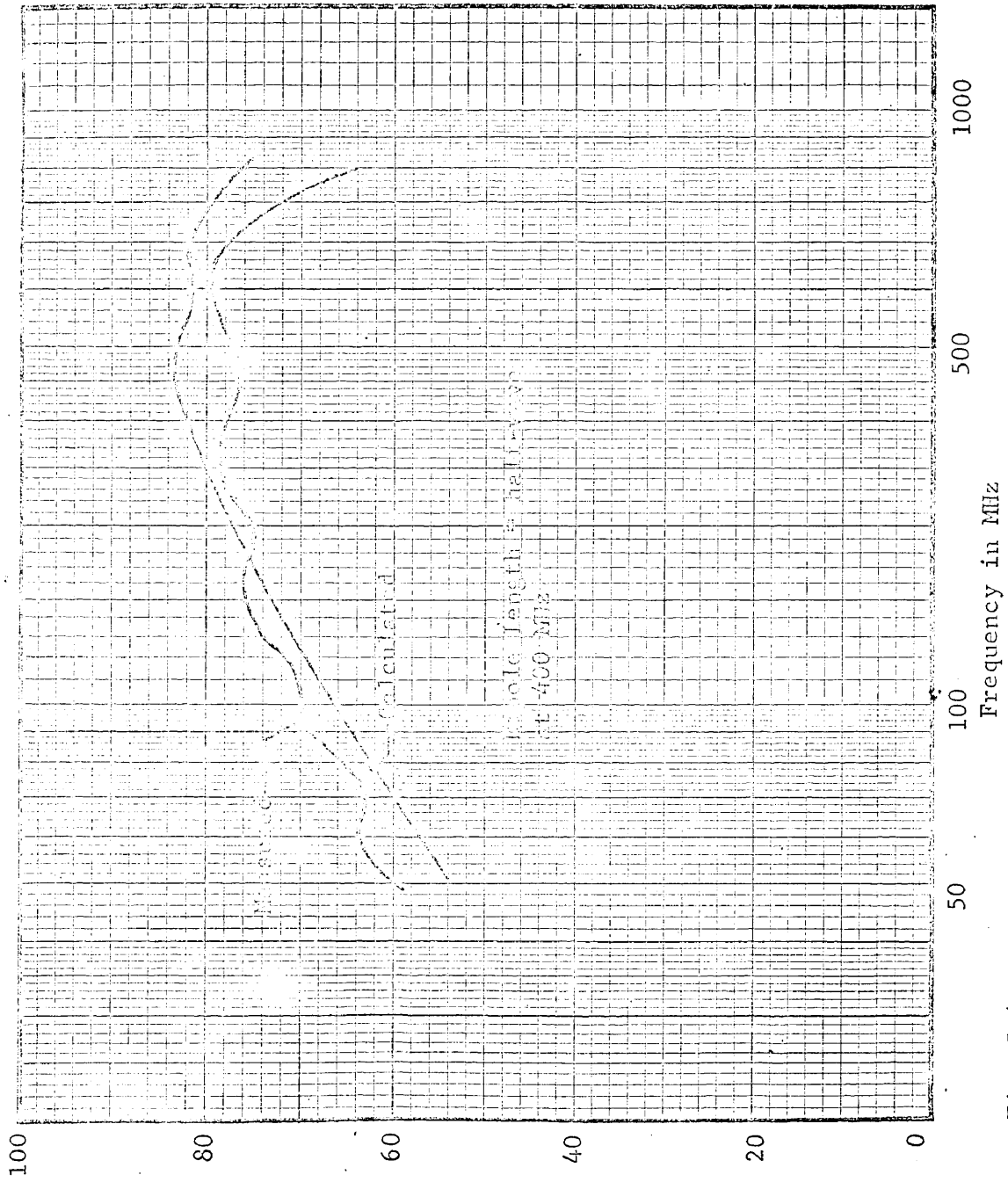


Fig. 5-4 Comparison of Measured and Calculated Electric Field Strengths.

In general, the measured values were slightly higher than the calculated values for frequencies below 400 MHz whereas the calculated values were higher for frequencies above 400 MHz. It is believed that the lack of agreement outlined above is due to the current distribution along the antenna being other than sinusoidal. Another reason for the lack of agreement could be that the driving source was a band of frequencies instead of a single sinusoid.

Relative Impedance Level of the Spark and the Antenna

The power transferred from a source to a load is a function of the impedance levels of each component. If a source is represented by an ideal current source shunted by a source admittance, one could calculate the current supplied to a load. If the load admittance is large compared to the source admittance, the current in the load will be almost independent of the load admittance. The current in the load will be nearly equal to that of the ideal current source.

One may obtain a qualitative relationship between the antenna impedance and the noise source impedance from the plots of the measured signal amplitude versus frequency and the known impedance at the center of a dipole.

The input impedance of an antenna is a complicated function of frequency which cannot be described in any simple analytical form [24]. For the dipole antenna, however, Jordan [25] shows by plots how the impedance at the center changes as a function of frequency. The impedance changes from a minimum of 70 ohms at the half-wave resonant frequency to 1000 ohms or more at the full wave resonant frequency. For the antenna length very short compared to a wave length, the impedance increases to several thousand ohms. The effect of the antenna impedance on the radiated signal as shown by Figure 5-2 is negligible because the amplitude of the electric field strength does not change rapidly with fast changing impedance levels.

The model, as described in Chapter IV, to represent the spark noise source was an ideal current source shunted by a 10^{-7} mho conductance and a 4 pF capacitance. Since the capacitance measured at the center of the dipole is a part of the antenna impedance, this component of the model will be assumed to be associated with the antenna. The remaining component, the 10^{-7} mho conductance, is very small compared to the admittance of the antenna, as described above, and may be neglected. The writer believes the noise source antenna system is approximated by an ideal current source driving a center fed dipole. The current distribution on the antenna is not known but is believed that it is not

sinusoidal.

Summary of Chapter V

This chapter showed the measured amplitude of the radiated electric field by plots and compared them to calculated values. The model chosen to represent the radiating system was shown to be a valid model with the spark acting like an ideal current source. Other characteristics of the radiated noise were:

1. For frequencies below 200 MHz the amplitude of the electric field increased 40 dB per decade for all lengths of antennas used in this research.
2. The radiated electric field was a maximum in the 300 to 600 MHz region and was approximately constant across this frequency range.
3. The spark as a broadband noise source may be represented as an ideal current source if the load impedance is no more than a few thousand ohms.

CHAPTER VI

CONCLUSIONS

This study revealed that the radiated electromagnetic noise produced by small dimension, low rms current sparks is composed of frequency components which extend well into the gigahertz region. The unexpectedly large peak current of several amperes which existed in a repetitive spark whose rms value was a few hundred microamperes gives one a clearer picture of the importance of this type of interference.

Although the power radiated by a spark is a function of the period, this study showed that the amplitude of the interference noise, when processed by a band-limited amplifier, is a function of the spark wave shape and amplitude but is independent of the period.

The radiated noise produced by a spark associated with a small dipole was found to increase at a rate of about 40 dB per decade until a broad peak was reached about 400 MHz. The noise was observed to decrease above about 600 MHz. This suggests that a noise detector operating in the 400 to 600 MHz range should be efficient because the noise amplitude would be relatively large in this frequency range and the corresponding Yagi directional-type antenna would be small.

This work was limited to small, repetitive sparks and dipole radiating systems. It is believed that many cases

of radio frequency interference (RFI) found on electric power lines are accurately portrayed by the model developed herein. However, other cases are found in significant numbers, cases such as the ground wire acting as a vertical monopole radiator or the long conductor performing as a long-wire traveling-wave antenna. Further investigation of the characteristics of the RFI produced by the latter two configurations should supply information about more suitable methods of locating this type RFI source.

BIBLIOGRAPHY

- [1] Leonard B. Loeb, Fundamental Processes of Electrical Discharge in Gases. New York: John Wiley and Sons, Inc., 1939, p. 413.
- [2] W. E. Pakala, E. R. Taylor, Jr., and R. T. Harrold, "High voltage power line siting criteria", Technical Report No. RADC-TR-66-607, p. 10, 1967.
- [3] Edward N. Skomal, "The dimensions of radio noise", IEEE Transactions on Electromagnetic Compatibility; Vol. EMC-11, p. 22, May, 1967.
- [4] John P. German, "Characteristics of electromagnetic radiation from gap-type spark discharges on electric power distribution lines", IEEE Transactions on Electromagnetic Compatibility, Vol. EMC-11, No. 2, p. 84, May, 1969.
- [5] Frederick A. Maxfield and R. Ralph Benedict, Theory of Gaseous Conduction and Electronics, New York: McGraw-Hill Book Co., Inc., p. 287, 1941.
- [6] John P. German, Op. Cit., p. 84.
- [7] Leonard P. Loeb, Op. Cit., p. 427.
- [8] Leonard P. Loeb, Op. Cit., p. 427.
- [9] Sir J. J. Thompson, Conduction of Electricity Through Gases, Vol. II., London: Cambridge at the University Press, p. 476, 1933.
- [10] Leonard B. Loeb, Op. Cit., p. 415.
- [11] Bureau of Ships: Navy Dept., "The Radio Interference Meter", Navships 94180, p. 15, July, 1962.
- [12] Mischa Schwartz, Information Transmission Modulation and Noise, New York: McGraw-Hill Book Co., Inc., p. 392, 395, 1959.
- [13] John P. German, "A study of electromagnetic radiation from electric power lines", A Report from the Texas A & M Research Foundation, p. 7, Aug., 1968.

- [14] Erwin Kreyszig, Advanced Engineering Mathematics, New York: John Wiley and Sons, Inc., p. 475, 1963.
- [15] H. H. Skilling, Electrical Engineering Circuits, New York: John Wiley and Sons, Inc., p. 431, 1960.
- [16] John D'Azzo and C. H. Houpis, Feedback Control System Analysis and Synthesis, New York: McGraw-Hill Book Co., Inc., p. 180, 1960.
- [17] Richard C. Dorf, Modern Control Systems, Reading, Pa.: Addison-Wesley Publishing Co., p. 183, 1967.
- [18] John D'Azzo and C. H. Houpis, Op. Cit., p. 178.
- [19] John D'Azzo and C. H. Houpis, Op. Cit., p. 179.
- [20] John D'Azzo and C. H. Houpis, Op. Cit., p. 181.
- [21] Edward C. Jordan, Electromagnetic Waves and Radiating Systems, Inglewood Cliffs, N. J.: Prentice Hall, Inc., p. 395, 1950.
- [22] John D. Kraus, Antennas, New York: John Wiley and Sons, Inc., p. 276, 1950.
- [23] Edward C. Jordan, Op. Cit., p. 395.
- [24] Edward C. Jordan, Op. Cit., p. 452.
- [25] Edward C. Jordan, Op. Cit., p. 482, 483.

APPENDIX A COMPUTER PROGRAM

This appendix contains the computer program written in Fortran IV language for use on the IBM 360-65 computer. This program computed the Fourier series components of the spark waveform using numerical integration. The trapezoidal rule was used to integrate because more accurate results were obtained with the trapezoidal rule than Simpson's rule. The accuracy of the program was determined by comparing the computed values of the amplitudes of the Fourier series components representing square and sawtooth waveforms with their calculated values.

The input data read in from cards consisted of the following:

P = Period of the spark in seconds

DELTA = Integration interval in seconds

Y(J) = Amplitude in volts of the time domain waveform
at the beginning of the Jth interval.

```

=====
CJOB W603T7,TIME= 2,PAGES= 20 W L SEASLEY FE DEPT
$1B5CX E.E. DEPARTMENT
C=====
C***** THIS PROGRAM WILL COMPUTE THE AMPLITUDE SPECTRUM OF THE FOURIER
C***** SERIES REPRESENTING A SHORT PERIODIC PULSE OF VARIOUS SHAPES AND PERIODS
C***** INTEGRATION IS PERFORMED USING THE TRAPEZOIDAL APPROXIMATION
C*****
C=====
19 WRITE(6,10)
18 FORMAT(1H1,7X,'FREQ IN HZ',3X,'P=0.01',5X,'P=0.001',4X,
C 'P=0.0001',3X,'P=0.00001',4X,'CIN DB ABOVE 1 UV')
C DIMENSION Y(57),A(57),B(57)
C READ(5,2) P, DELTA
2 FORMAT(2F8.2)
IF (P.EQ.100.) GO TO 30
READ(5,1)(Y(J),J=1,57)
1 FORMAT(10 F8.2)
C Y(1) IS THE VALUE OF THE TIME FUNCTION AT START OF ITH INTERVAL
C A(I) WILL BE THE VALUE OF Y(I) TIMES COS(W*I)
C B(I) WILL BE THE VALUE OF Y(I) TIMES SIN(W*I)
C P IS THE PERIOD OF THE PULSE, DELTA IS THE TIME INTERVAL BETWEEN
C Y(I) AND Y(I+1)
C F= FUNDAMENTAL FREQUENCY IN HZ
DO 12 K=6,8
IF(K.EQ.7) GO TO 14
1=10
=====

```

```

14 GO TO 15
15 L=1
16 DO 12 J=10,90,1
17 IF (J.EQ.51) GO TO 19
18 AJ=10*J
19 DO 12 I=2,10,2
20 R=1*10**7
21 X=AJ*10.**K +R
22 T=0.
23 S = 6.2832*X
24 DO 6 N=1,57
25 X IS THE FREQUENCY OF THE HARMONIC
26 T IS THE VALUE OF TIME CORRESPONDING TO Y(N)
27 A(N)=COS(S*T)*Y(N)
28 B(N)=SIN(S*T)*Y(N)
29 T = T + DELTA
30 SUMA = 0.
31 SUMB = 0.
32 DO 7 M= 2,56
33 SUMA = SUMA + A(R)
34 SUMB = SUMB + B(R)
35 COEFA = (DELTA )*(A(1)+ 2. * SUMA + A(57))/P
36 COEFB = (DELTA )*(B(1)+ 2. * SUMB + B(57))/P
37 C2 = SQRT(COEFA**2 + COEFB**2)
38 C5=20.*ALOG10(C2*1.0E 06)
39 C2 IS THE AMPLITUDE OF THE HARMONIC WHOSE FREQ IS X
40 THE NUMBER OF THE HARMONIC IS X DIVIDED BY F(0)
41 C1=0.
42 C3 = 0.
43 C4 = 0.

```

```
12 WRITE (6,8) X,C1, C2, C3, C4 ,C5  
8 FORMAT (1H,4X,E11.3, 4E12.3,E17.3)  
GO TO 19
```

```
30 STOP  
END
```

```
$DATA
```

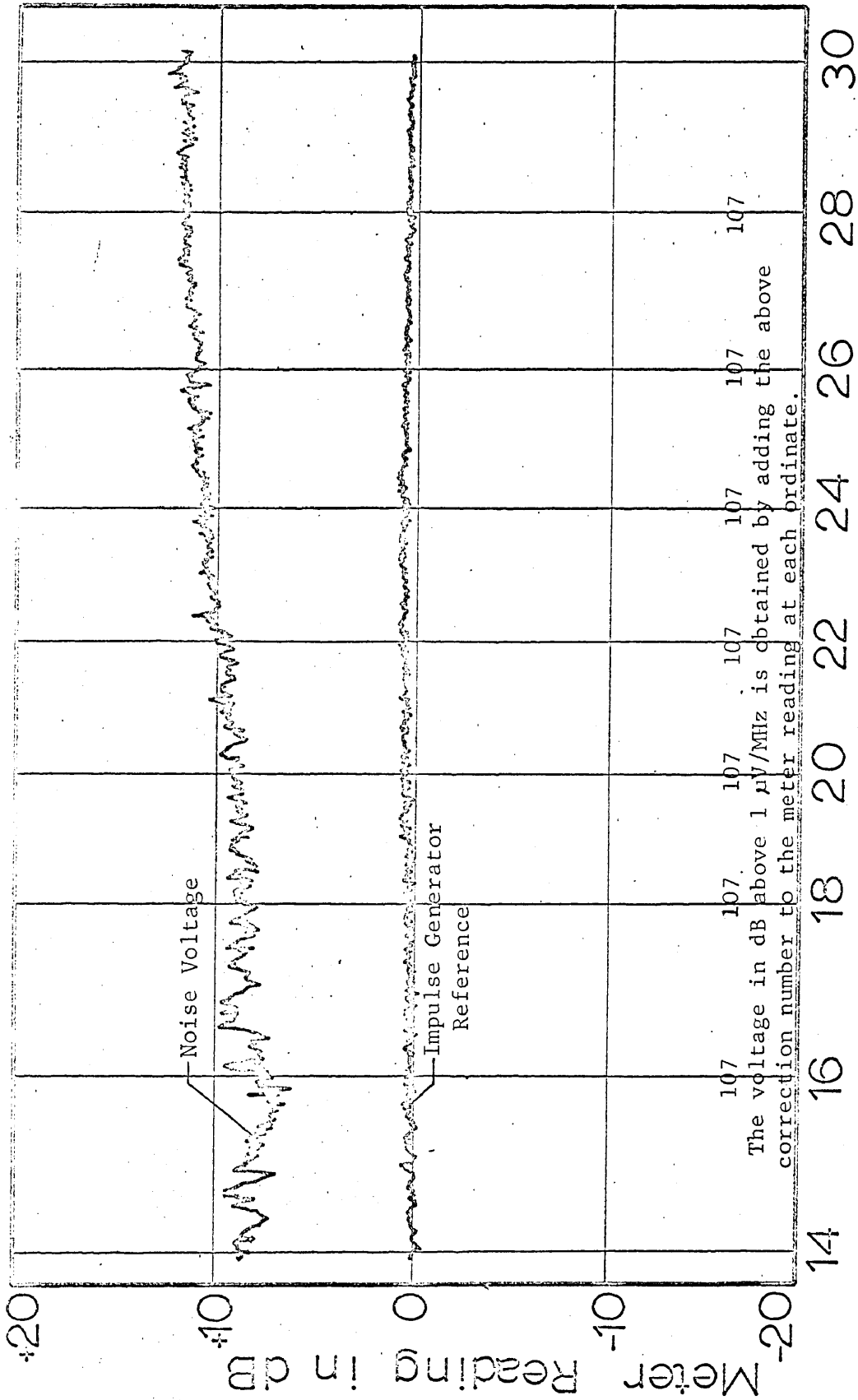
APPENDIX B MEASURED VALUES

Appendix B contains a series of 15 plots of the measured frequency domain characteristics of the spark current. Approximately one octave of frequency is covered on each plot with the total frequency span being from 14 kilohertz to one gigahertz. The units of the ordinate are meter readings in dB above one microvolt per megahertz.

The plots were obtained by means of an EMC-25 Noise Detector supplying voltages suitable for driving an x-y plotter. The input to the EMC-25 was obtained from a 50 ohm resistor through which the spark current existed.

The EMC-25 was calibrated at the low frequency end of each band by means of an internal impulse generator acting as a reference and a plot of the reference signal placed on each figure. The change in the recorded value of the reference signal from its value at the calibration point (the start) was then used as one factor in calculating the amplitude of the spark noise voltage at other frequencies. The other factors used to calculate the amplitude of the noise voltage were the EMC-25 attenuator setting, the broad-band correction factor (supplied by the manufacturer), and the value of the attenuation inserted in the coaxial line between the EMC-25 and the noise source.

The amplitude, at a particular frequency, was then calculated by adding algebraically the EMC-25 meter reading to the following correction factors: the change in EMC-25 gain as outlined above, the attenuator setting, the external attenuation and the broad-band correction factor. Since the EMC-25 reads rms value of an equivalent sinusoid, the peak value of the measured voltage is obtained by adding 3 dB. All of these correction factors are added and their sum is placed at the bottom of each ordinate on Figures B-1 through B-15.



Frequency, kHz Band 1

Fig. B-1 Meter Reading versus Frequency for a 0.030 Inch Gap.

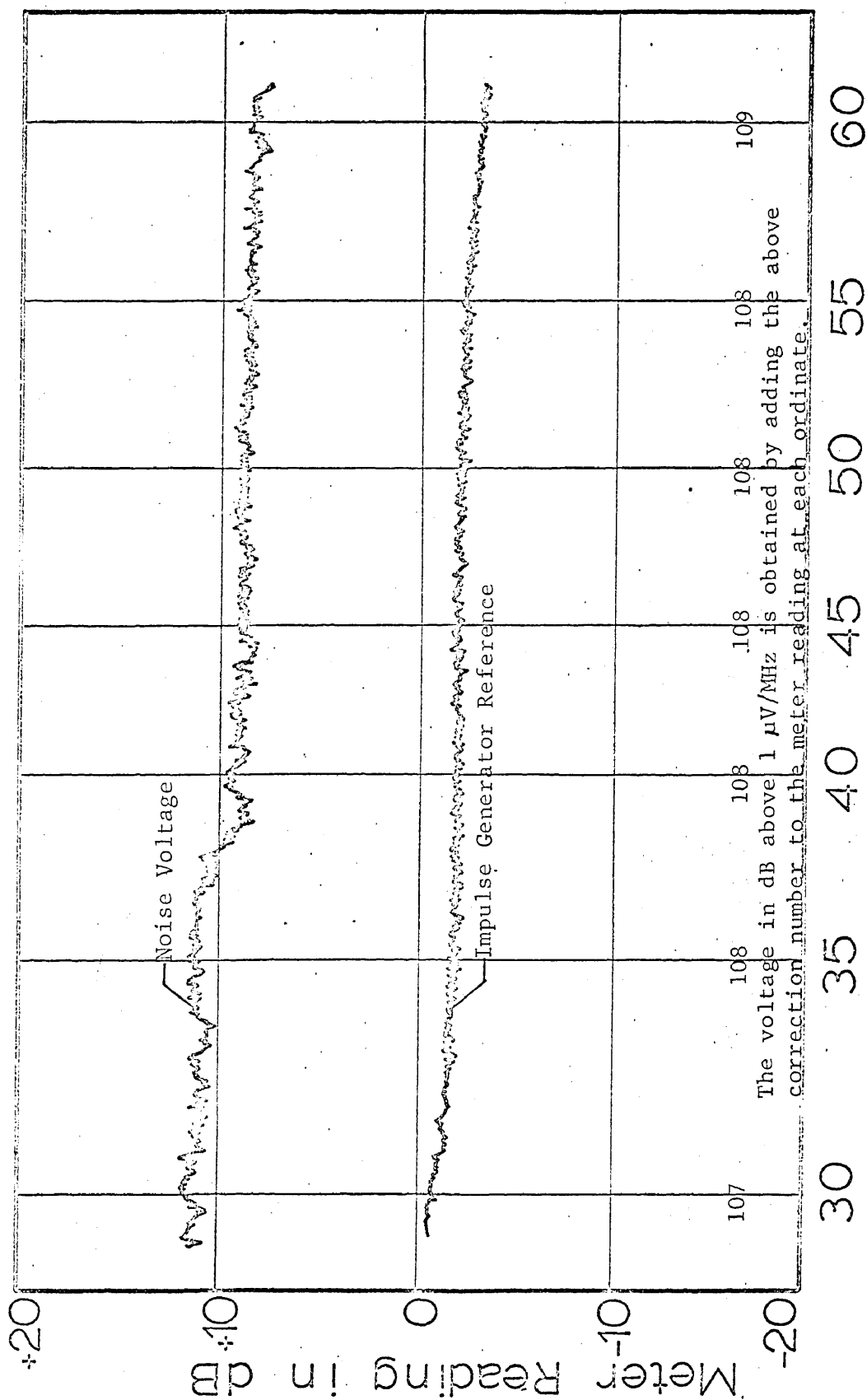


Fig. B-2 Meter Reading versus Frequency for a 0.030 Inch Gap.

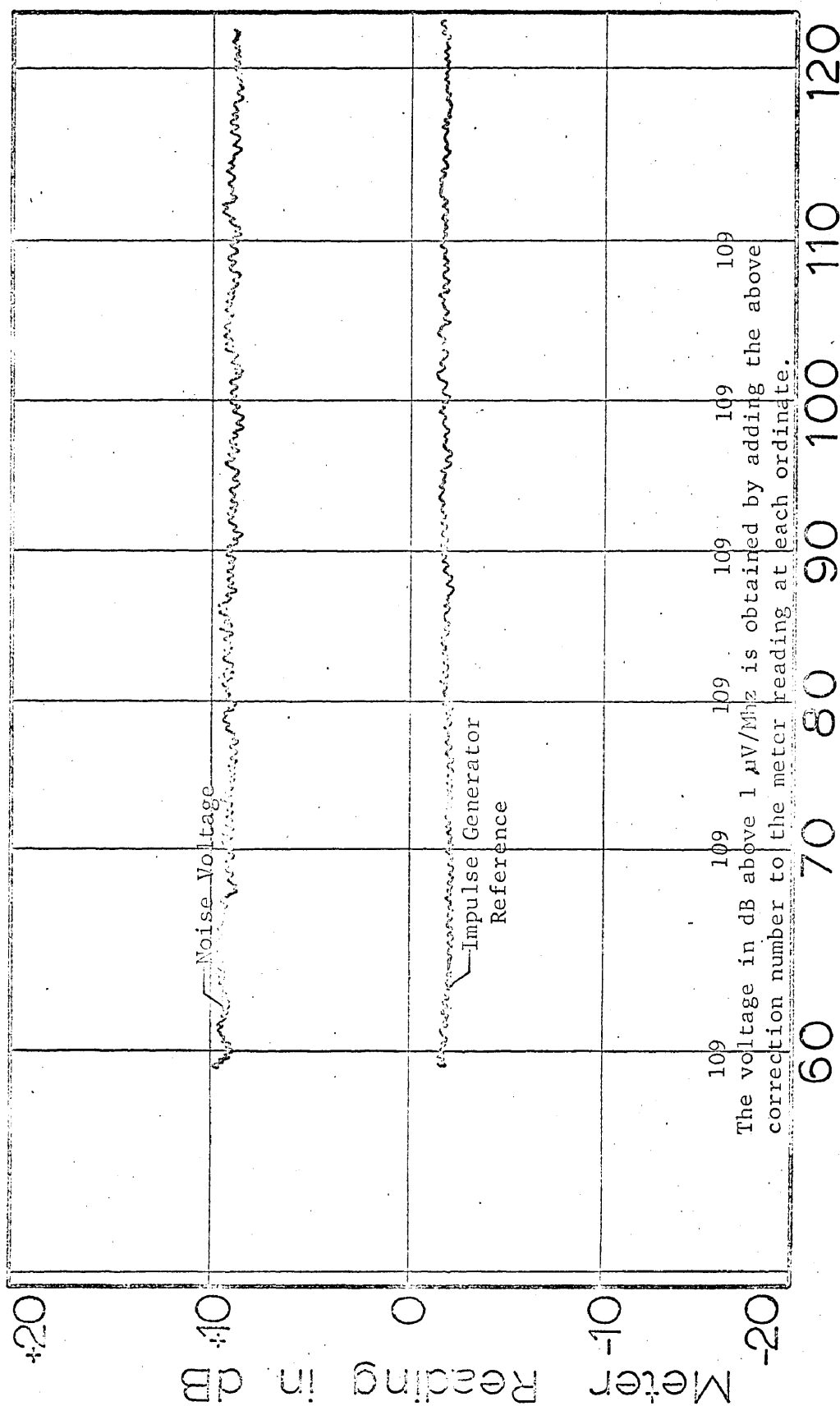
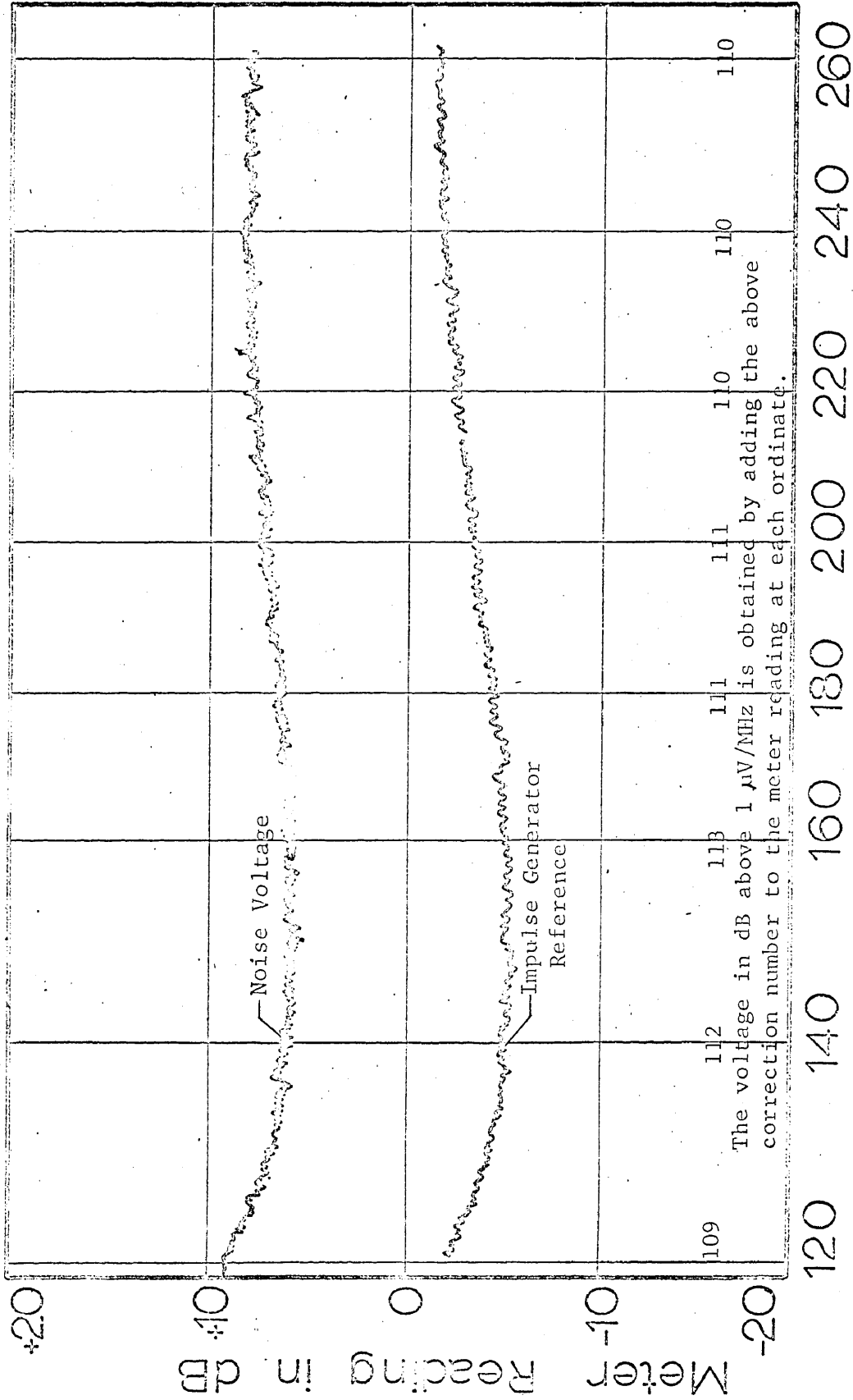
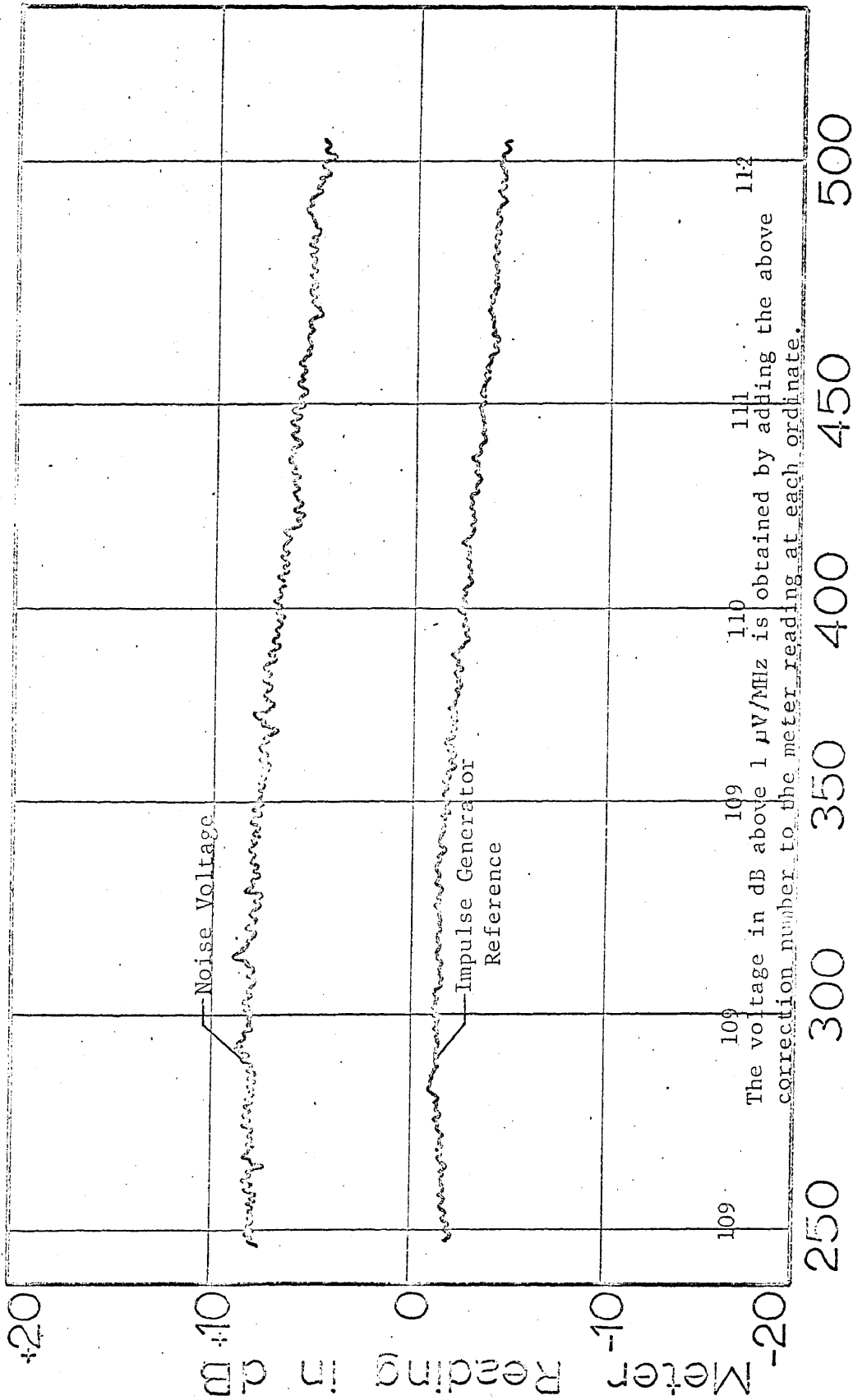


Fig. B-3 Meter Reading versus Frequency for a 0.030 Inch Gap.



Frequency, kHz Band 4

Fig. B-4 Meter Reading versus Frequency for a 0.030 Inch Gap.



Frequency, kHz Band 5

Fig. B-5 Meter Reading versus Frequency for a 0.030 Inch Gap.

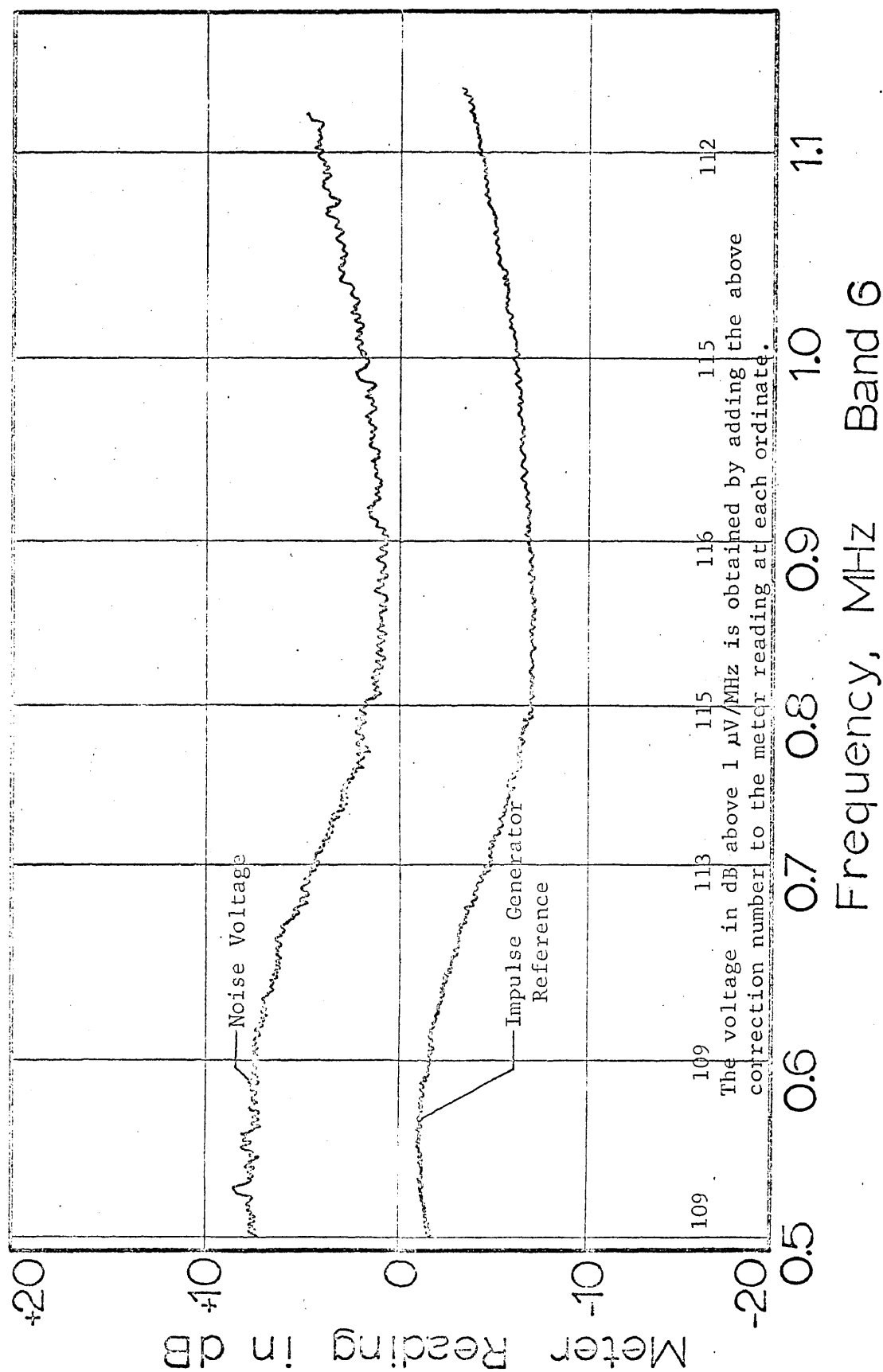


Fig. B-6 Meter Reading versus Frequency for a 0.030 Inch Gap.

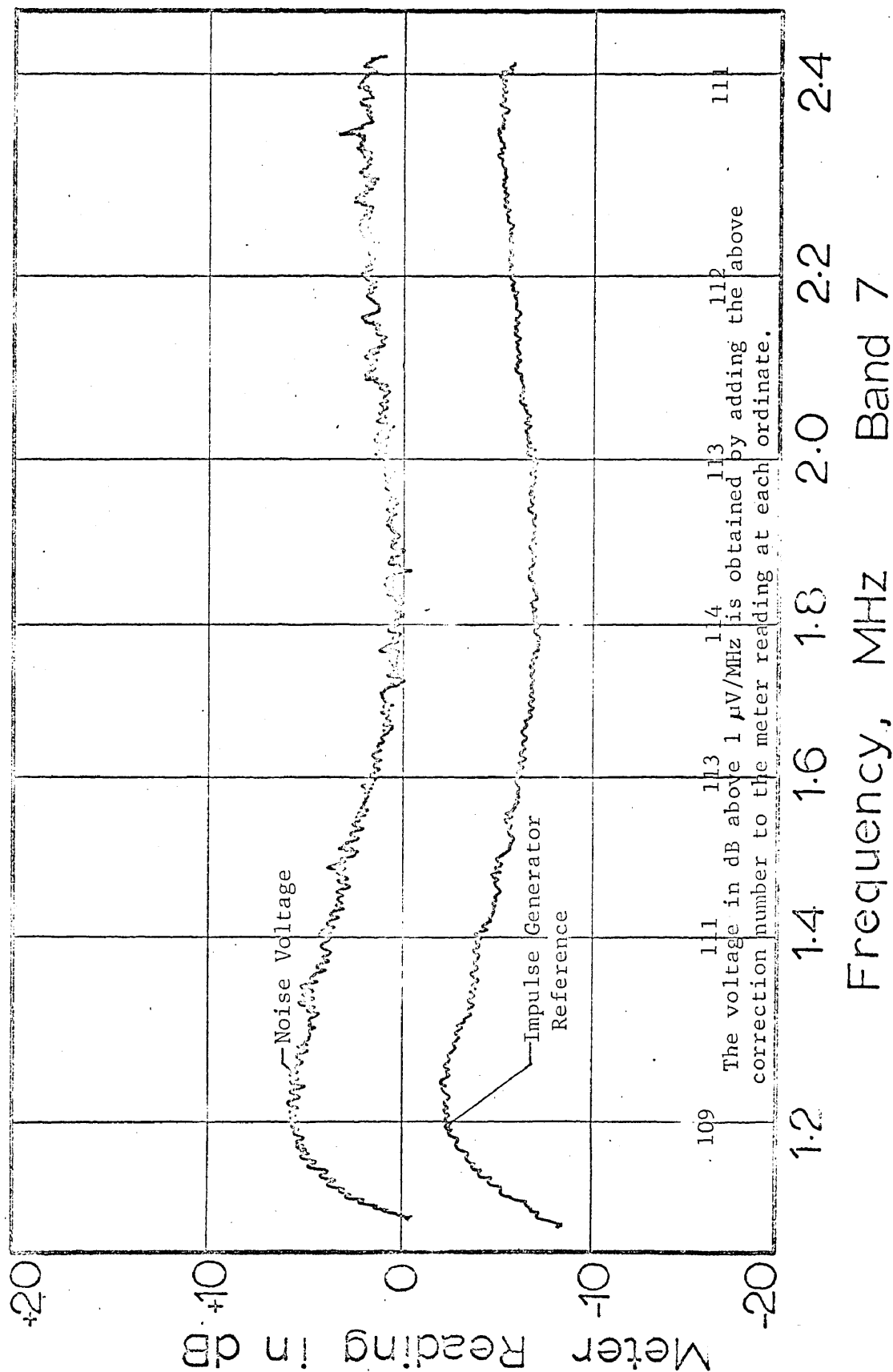


Fig. B-7 Meter Reading versus Frequency for a 0.030 Inch Gap.

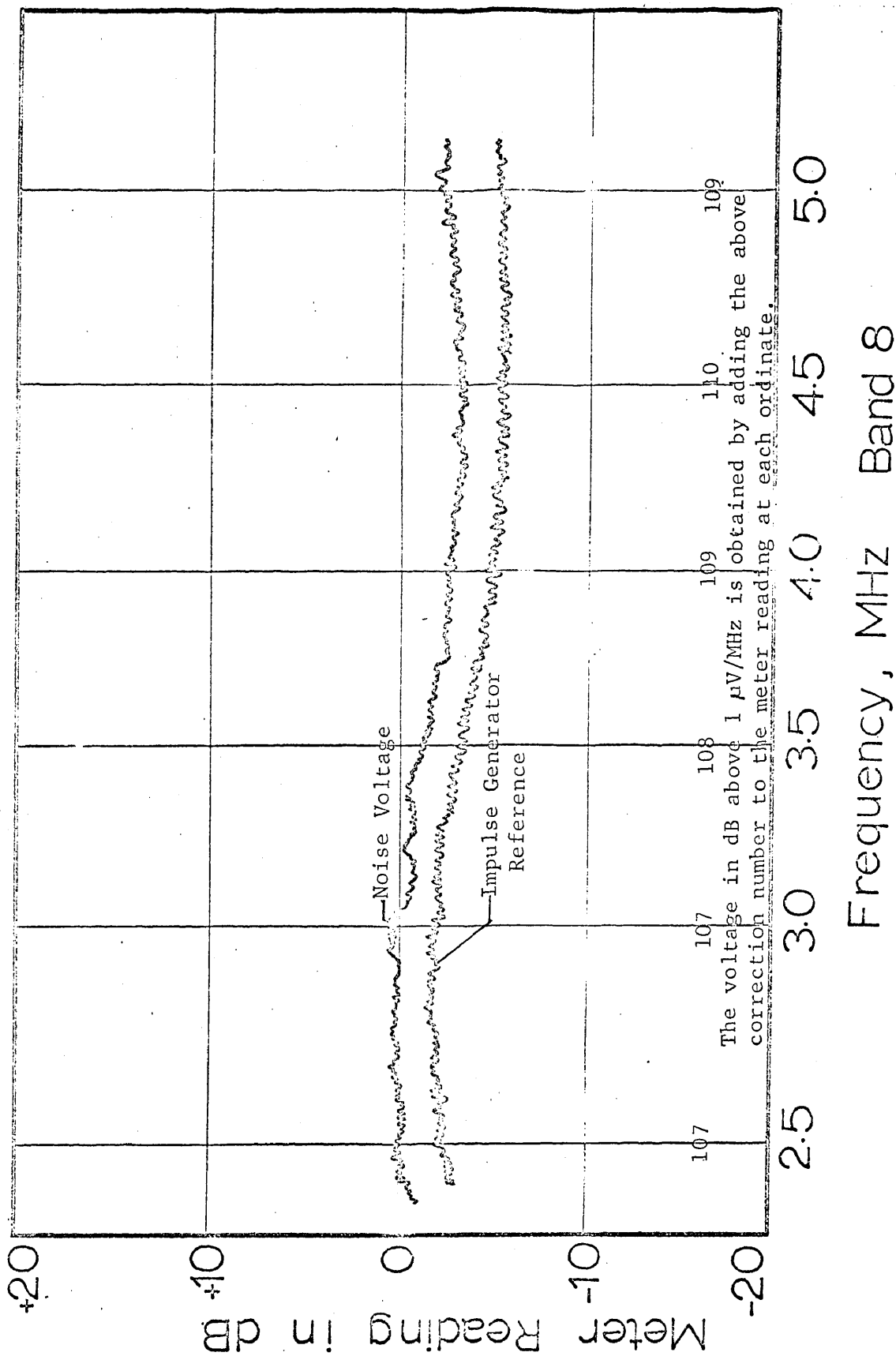


Fig. B-8 Meter Reading versus Frequency for a 0.030 Inch Gap.

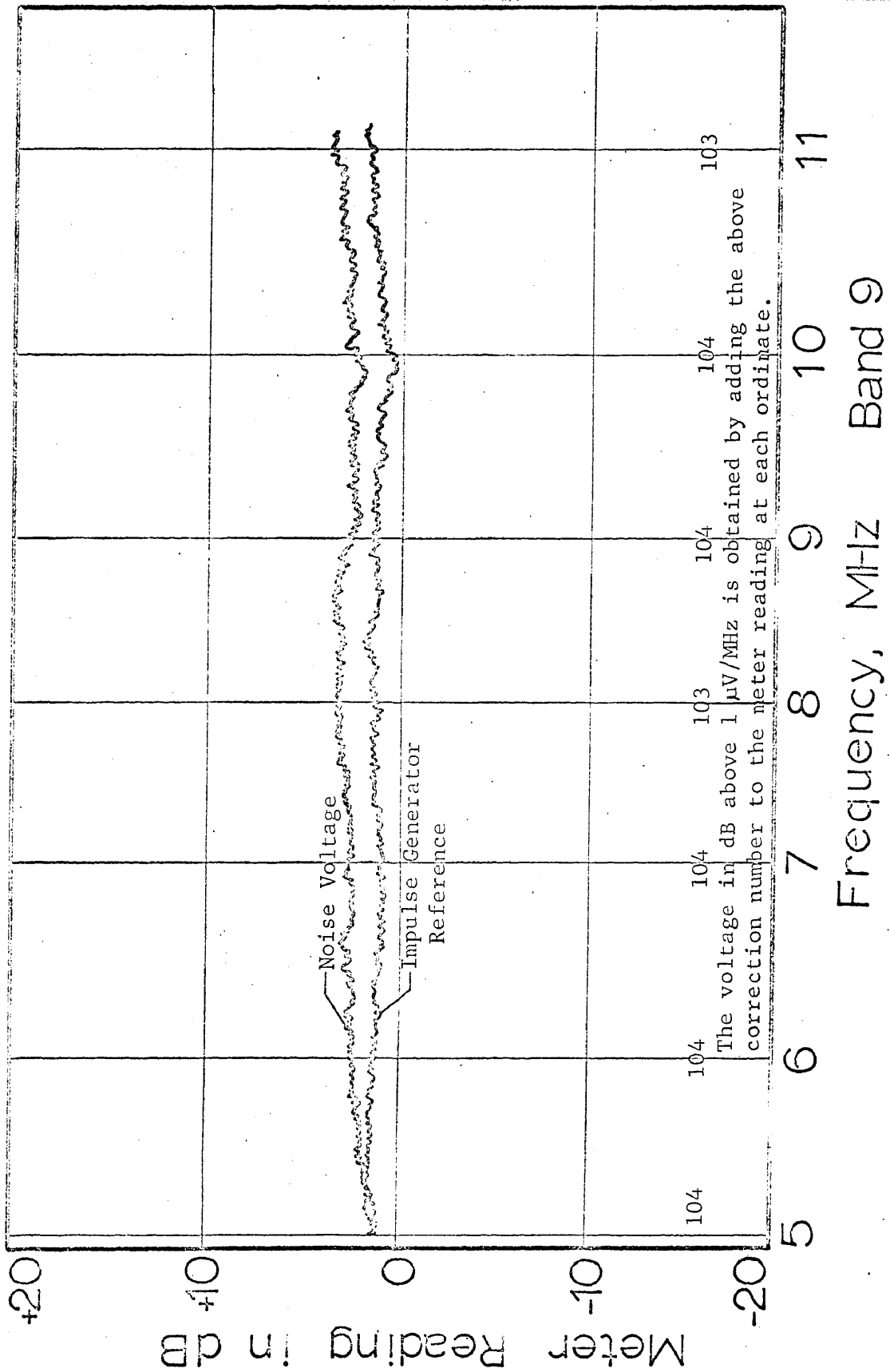


Fig. B-9 Meter Reading versus Frequency for a 0.030 Inch Gap.

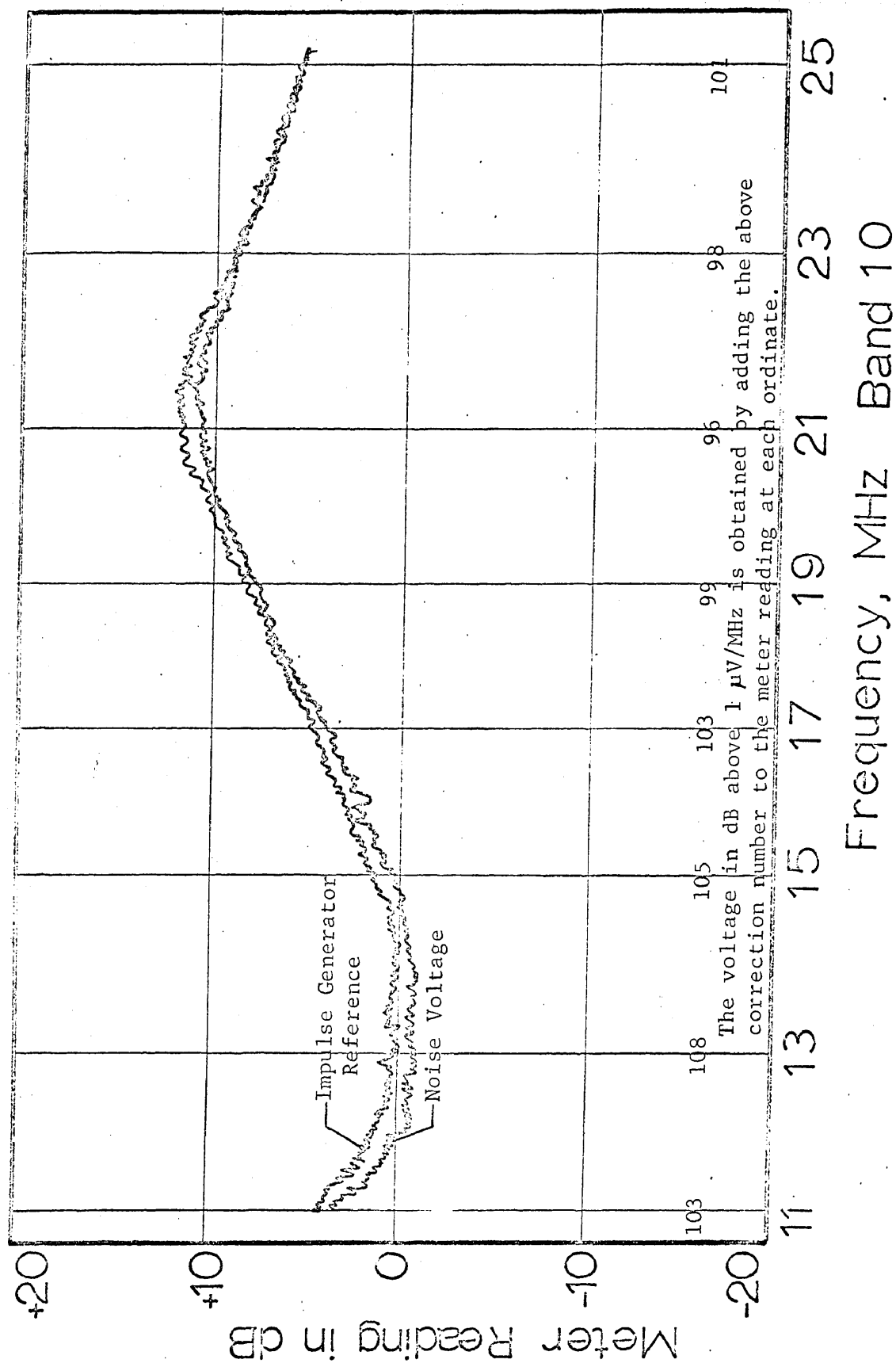


Fig. B-10 Meter Reading versus Frequency for a 0.030 Inch Gap.

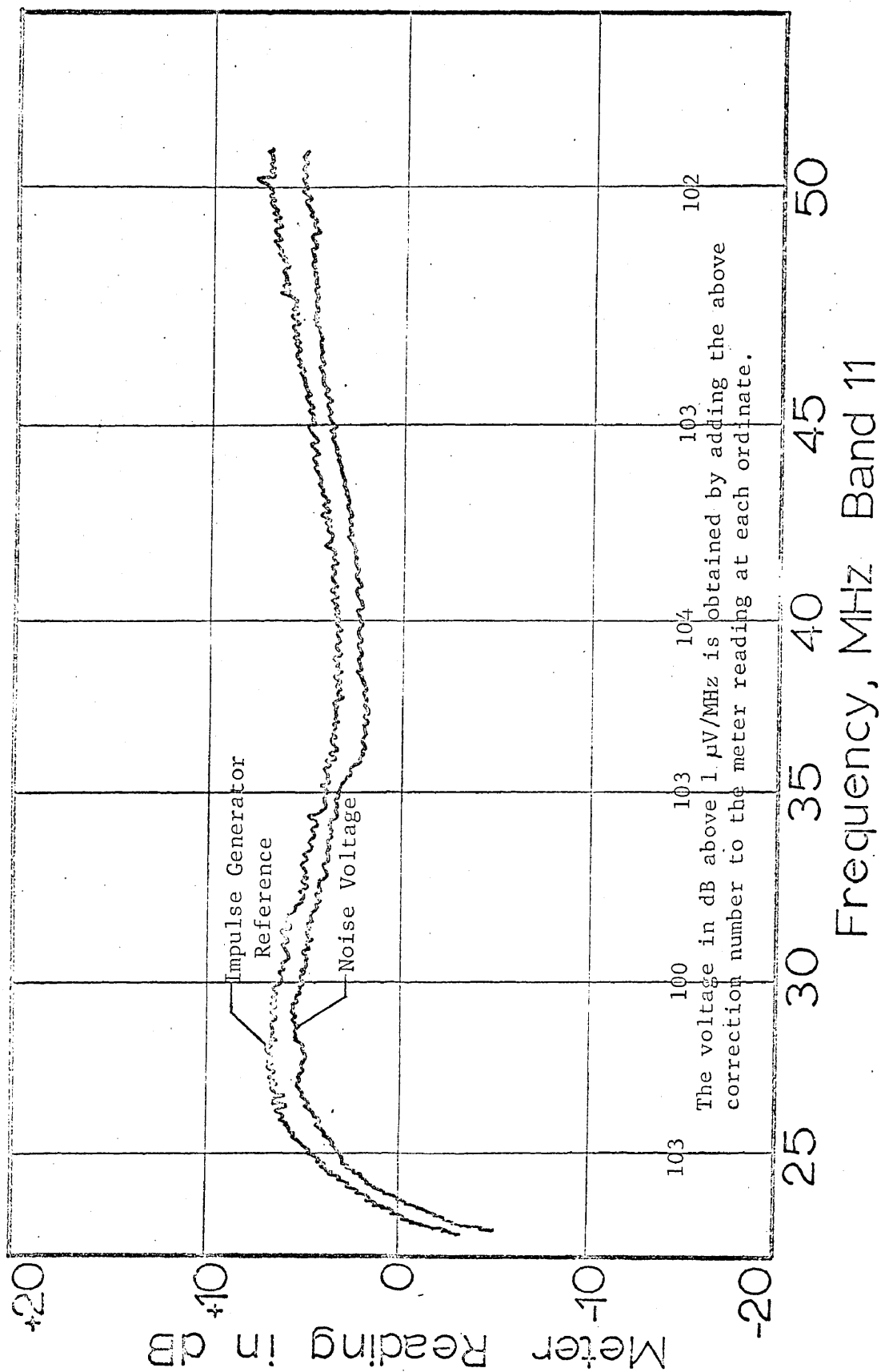


Fig. B-11 Meter Reading versus Frequency for a 0.030 Inch Gap.

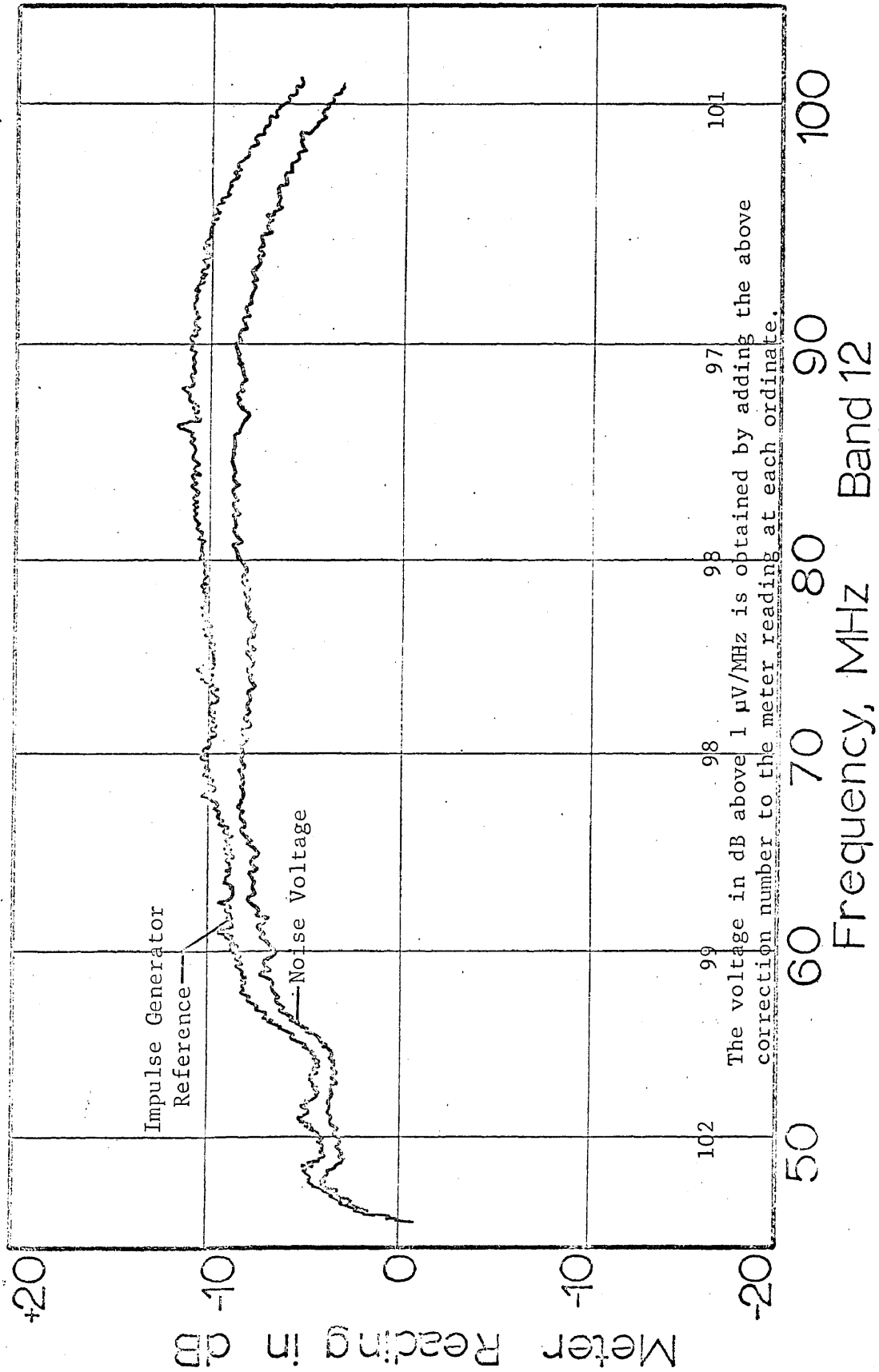


Fig. B-12 Meter Reading Versus Frequency for a 0.030 Inch Gap.

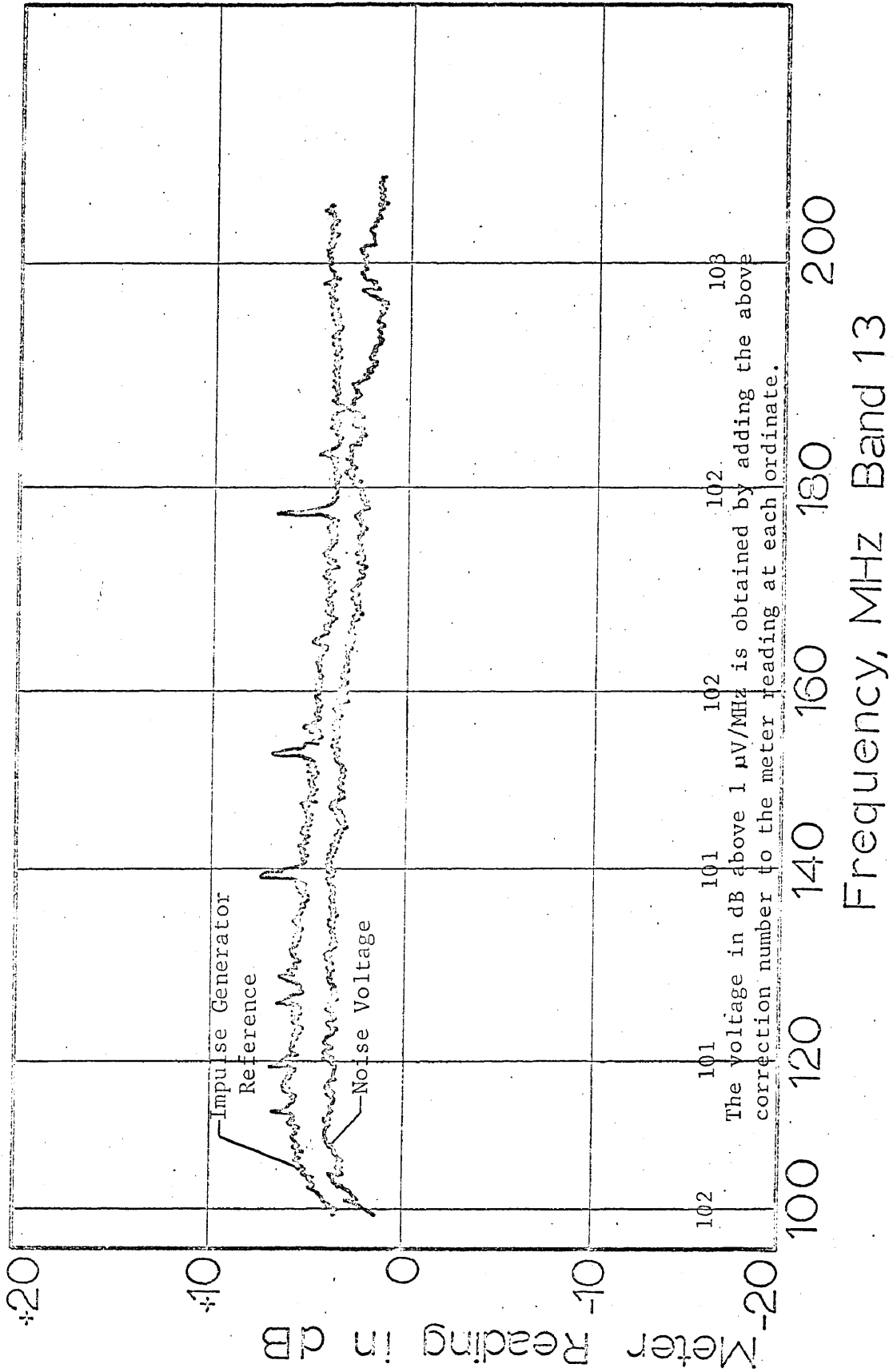


Fig. B-13 Meter Reading versus Frequency for a 0.030 Inch Gap.

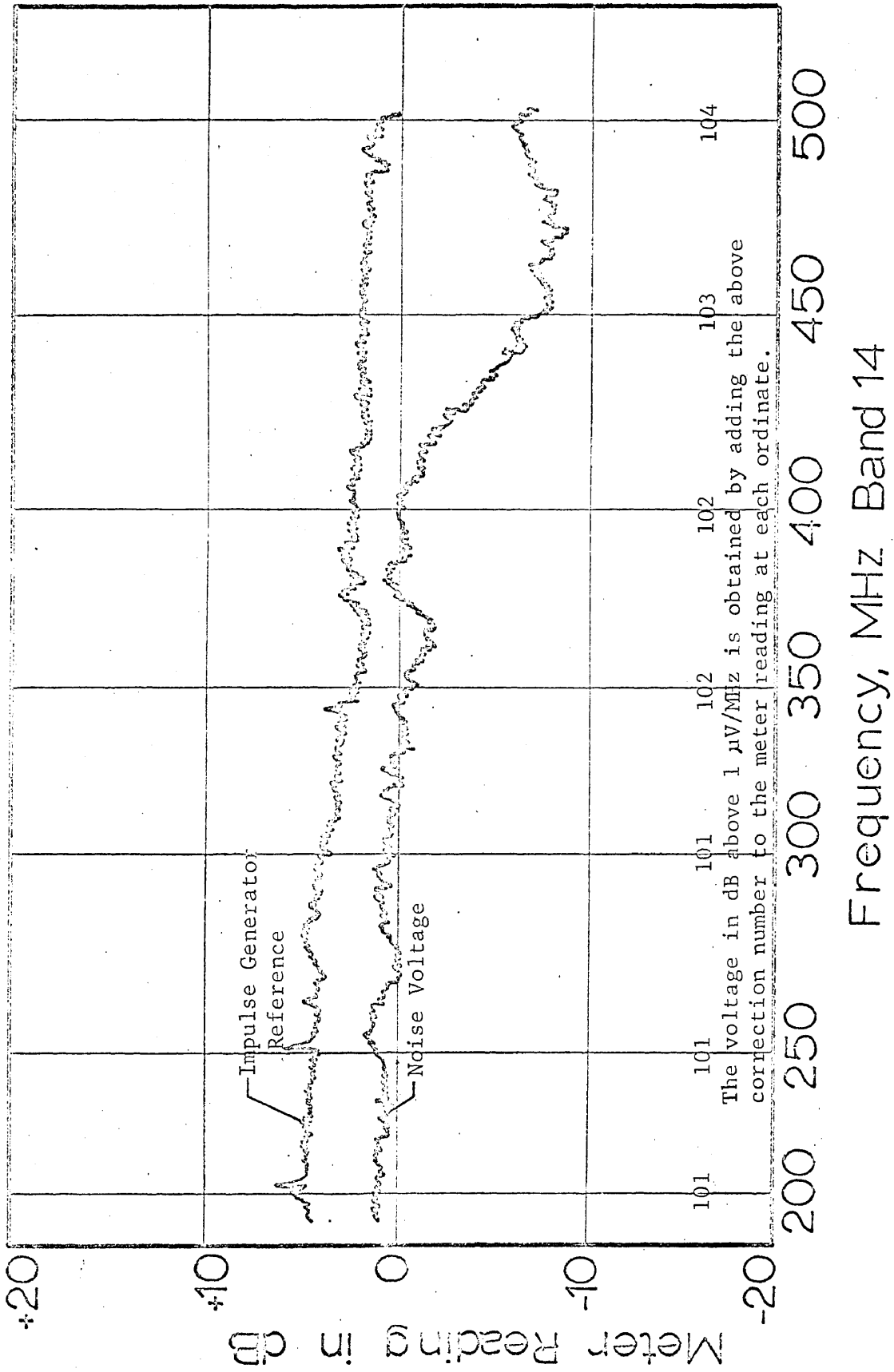


Fig. B-14 Meter Reading versus Frequency for a 0.030 Inch Gap.

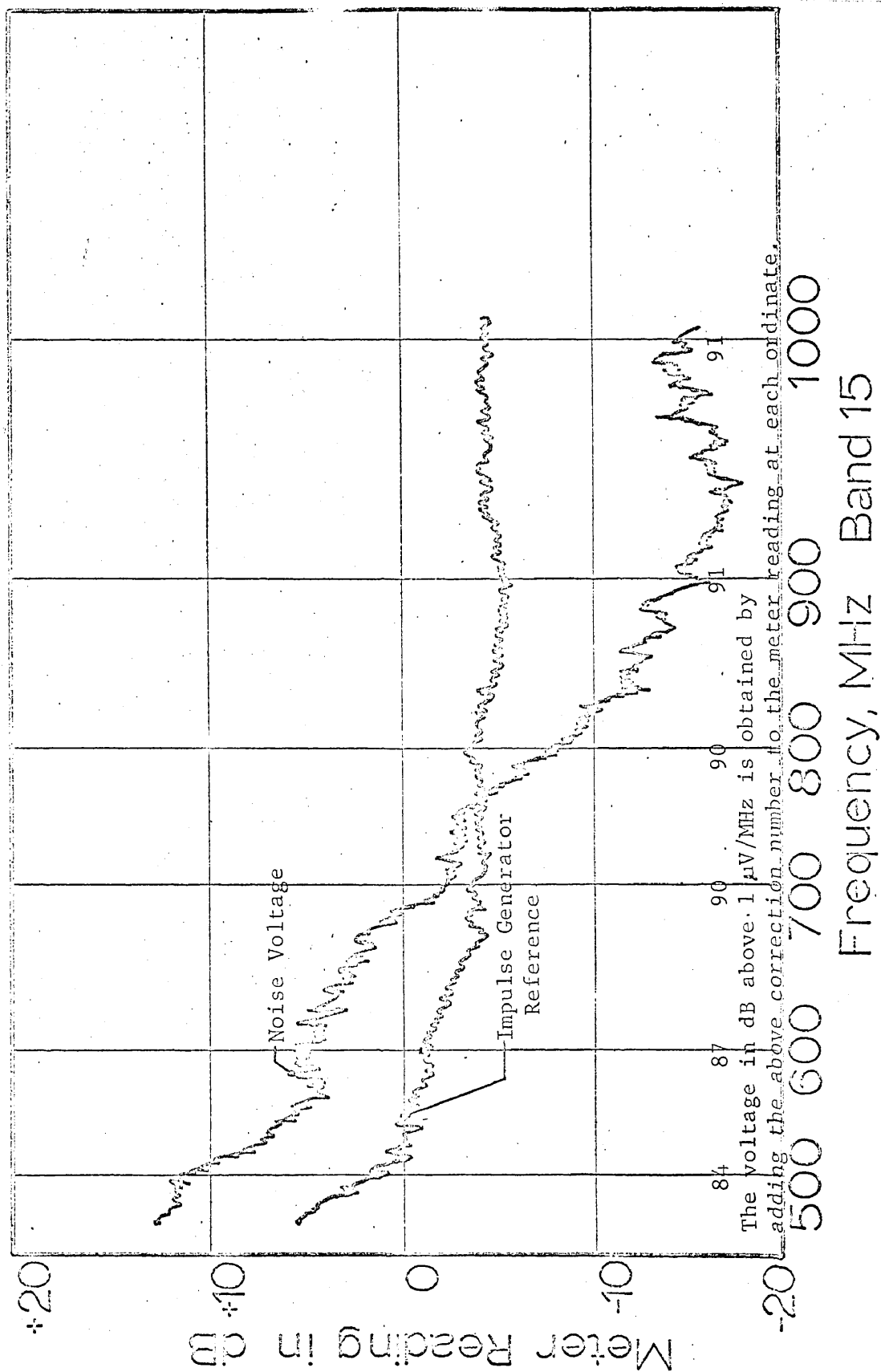


Fig. B-15 Meter Reading versus Frequency for a 0.030 Inch Gap.